

Received 26 July 2025, accepted 18 August 2025, date of publication 25 August 2025, date of current version 3 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3602476

RESEARCH ARTICLE

Self-Supervised Learning Graphical Neural Network Driven Prediction Model for Path-Planning and Navigation in Smart Sustainable Agriculture

R. NIVETHA¹, K. C. SRIHARIPRIYA¹, AND BALAMURUGAN BALUSAMY²

¹School of Electronics Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India

²Computer Science and Engineering, Shiv Nadar Institution of Eminence Deemed to be University, New Delhi 201314, India

Corresponding author: K. C. Sriharipriya (sriharipriya.kc@vit.ac.in)

ABSTRACT Recently, autonomous systems in agriculture have garnered increased attention among research communities and industries due to their greater significance. With rapid technological advances, numerous autonomous systems have been actively deployed in agriculture to accomplish complex tasks, such as surveying agricultural fields, managing the water cycle, resolving soil compaction issues, etc. Swarm robotics, path planning, and task assignment systems play a crucial role in agricultural automation and pave the way for smart, sustainable agriculture. However, the major problem here is that this system depends on dynamic path planning activities, which are the most crucial task for autonomous robots. This is because in smart agriculture, multiple agents are associated with the system to perform various activities on their way to the destination, making the path planning actions more complicated than ever. To address these constraints, this paper presents a cost-efficient, reliable, and safe navigable swarm approach, called the Self-Supervised Learning Graphical Neural Network Driven Prediction Model (SGNN), for smart sustainable agriculture. The proposed SGNN approach enables swarm robots to select the most optimized paths, thereby reducing the distance travelled by the robots. The work's prime focus is on developing a forecasting model for the efficient navigation and landing of swarm robots in innovative agricultural systems. The Semantic Drone Dataset is used for experimentation purposes. It is observed from the analysis that the proposed approach offers more than 85% percent accurate results in comparison to standard ResNet architectures. Additionally, this approach yields a score of around 0.8, indicating a good balance between precision and recall measures. Further, the result suggests a correlation coefficient factor of around 0.685, representing the effective classification of the target classes. Thus, the proposed SGNN approach has been experimentally proven to be more robust and efficient than existing approaches.

INDEX TERMS Swarm robots, sustainable agriculture, graph neural networks, self-supervised learning.

I. INTRODUCTION

In recent times, uncrewed aerial vehicles (UAVs), also known as drones, have gained popularity due to their viable and cost-efficient solutions to various problems. When it comes to autonomous systems, path planning capabilities play a significant role for different reasons. An efficient

path-planning algorithm is vital for autonomous control systems, particularly in IoT-based agricultural systems, where considerable automation and control are required to achieve the intended objectives [1], [2]. Energy-efficient and feasible path-planning algorithms are needed to avoid collisions the most. In this case, drones form the most efficient solution as they offer a safer alternative to human assistance and have been widely used in various applications. Following the advancement in technology and decreased production

The associate editor coordinating the review of this manuscript and approving it for publication was Mu-Yen Chen¹.

costs, there is currently an increased interest in developing drones for commercial applications. The increased versatility of drones, as measured by various metrics, makes them an ideal solution for rescue-related operations, search and rescue missions, inspections, precision agriculture, and many other applications [3], [4].

In precision agriculture, drones fly above agricultural fields, capturing more detailed images and videos to detect pests and insects, identify crop health, and monitor water and nutrient levels, among other applications. Furthermore, they are widely used for automating more complex agricultural tasks that do not require human involvement. Some of the advantages of implementing drones in agriculture include improved productivity, increased crop management, decreased labour efforts, environmental benefits, and earlier detection of pests and insects [5], [6].

However, the effective implementation of drones in precision agriculture poses the challenge of path planning and navigation. The path selected by drones or robots should be collision-resistant with the surrounding objects. Path planning in UAV systems should be considered from three different aspects: motion planning, trajectory planning, and navigation. Path planning is related to collision-free movement of robots, whereas motion planning focuses on constraints such as turning a crank and flight path in path planning. Identifying the path with the shortest path length and minimum turn angle yields the best solution. In contrast, navigation covers all the aspects of trajectory planning, collision avoidance, localisation, and motion planning. It deals with the movement of drones from one place to another [7], [8].

When dealing with path planning-related issues in precision agriculture, 3-dimensional views of the complex environments have become the most vital component. The 2-dimensional views offer less clarity than the 3-D view. Thus, effectively segmenting the 3-D opinions obtained from the agricultural field will offer the most efficient path planning and navigation facilities. However, in the present and past, several research proposals have been discussed to address various path planning and navigation-related problems in UAVs [9], [10]. These proposals addressed the challenges of UAV-based precision agricultural systems in terms of multiple shapes and complexity measures. However, the primary objective behind all these approaches is to identify the most efficient and optimal path with reduced complexity and increased accuracy. The path generated by the algorithm should consume less time, use minimal energy, and reduce the effect of collisions among the drones [11], [12].

Furthermore, the algorithm should also satisfy the properties of robustness and completeness. Some challenges associated with efficient path-planning algorithms include optimality, accurate path planning, completeness, cost efficiency, time efficiency, robustness, energy efficiency, and collision avoidance [13], [14], [15].

Further, integrating advanced technologies in agriculture is crucial for enhancing productivity and sustainability.

According to the Food and Agriculture Organisation, global food production must increase by 70% by 2050 to meet the projected needs of a population of 9.7 billion. Graph Neural Networks (GNNs) have emerged as a powerful tool for modelling complex relationships in agricultural data, enabling more effective path planning and navigation. By leveraging self-supervised Learning, which reduces reliance on labelled datasets, this project aims to develop a prediction model that optimizes resource utilization and minimises environmental impact, ultimately contributing to innovative, sustainable agriculture practices and improving overall efficiency. These parameters form the prime focus of this proposed work, which are listed in detail in the forthcoming subsection.

A. RESEARCH OBJECTIVES

1. The key objective of this work is to enhance the features discussed above with more efficient self-supervised GNN-based algorithm measures. Here, the aim is to perform semantic segmentation of real-time agricultural images using a stage GNN algorithm, which will be described in detail in Section III of the paper. Semantic segmentation generally involves understanding visual scenes and can be viewed as a dense classification problem.
2. Another prime objective here is to predict the category label associated with every pixel in the input image. Similar to the classification task, the set of categories for regions of tasks should be predefined at an earlier stage and remain fixed. However, when moving towards drone-based solutions for precision agriculture. The need for more efficient AI algorithms has become crucial. This algorithm plays a significant role in path planning and navigation-related activities in drones.

B. RESEARCH CHALLENGES

There exist two notable challenges that complicate the implementation of these algorithms.

- The first issue is the scarcity of large datasets related to UAVs in agriculture. Although fewer datasets are available, the possibility of acquiring labelled datasets is extremely low. Furthermore, manually labelling these datasets is a time-consuming task.
- Next, the standard practice in navigation and path planning-related approaches is working with the region of interest, encompassing various obstacle detection classes with arbitrary sizes and shapes. Some examples of target classes include grass, people, stones, and vehicles. One typical solution to this problem involves aggregating representations into a fixed-sized patch, which is then cropped to form regions that provide region-level representations.

However, the primary concern is that conventional approaches fail to utilize the rich contextual information from real-time agricultural fields sufficiently. To effectively address this limitation, this paper presents a novel method for path planning and navigation in precision agriculture using

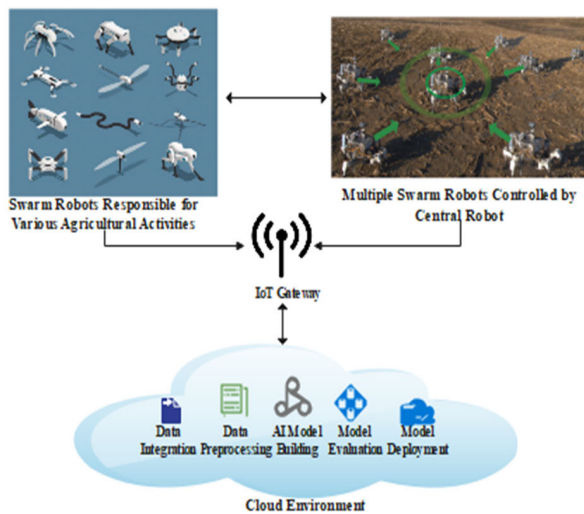


FIGURE 1. Magnetisation overview of the proposed architecture.

self-supervised GNN-based algorithms. The generic architecture associated with the proposed approach is described in detail in Figure 1.

The proposed SGNN approach involves a drone swarm that assists farmers in precision agriculture, enhancing their crop yields and mapping weeds more efficiently. Another set of swarm robots will be responsible for removing weeds and performing other agricultural-related tasks efficiently and simply without human intervention. The drones effectively pinpoint weed-infested areas, inspect them, and remove them promptly, enhancing crop production. The central swarm robot controls the swarm robots flying across the fields. Thus, the central robot controls and maintains the activity of various other swarm robots associated with it. A practical GNN-based self-supervised approach is proposed in this paper to segment images in the agricultural field and assist robots in their movement and landing processes. Only the generic architecture associated with the proposed algorithm is presented in this section, and a detailed explanation of its working is provided in the proposed work.

The paper is organised as follows: This section briefly introduces various challenges related to the implementation of drones in precision agriculture. Also, it provides a generic architecture associated with the proposed algorithm. Section II provides a brief review of the literature. Section III presents the proposed GNN algorithm in detail. Section IV presents the result analysis more detailedly, incorporating real-time observation and comparison with standard algorithms. Section V concludes the proposed approach. For a conference, please observe the conference page limits.

II. RELATED WORKS

This section summarises the literature related to the proposed SGNN method, highlighting its advantages and disadvantages.

An efficient approach for autonomous weed monitoring in precision agriculture is given in [16]. This paper presents

a cowbot, an automated weed-mowing robot designed for maintaining cow pastures. This work aims to enhance path planning in cow-bots, as the distribution of weeds on pastures remains unpredictable. Autonomous mowers with field and curvature views of constraints are analyzed in this algorithm. This work presents two major algorithms: the first for weed detection, and the second for path planning and coverage. The experiment shows that this algorithm leads to a 60% reduction in path lengths compared to conventional methods. The two algorithms used in the analysis process are random-search-based coverage paths and boustrophedon.

Another approach for robot navigation and obstacle detection using reactive neural networks is given in [17]. The primary focus of this work is to crop, rescale, and rotate the obstacle map according to the specified requirements. The obstacles are identified based on the agent's current and goal locations. These instances will assist in better map representation and correlation concerning the general navigation tasks. In addition to the obstacle map, an input map of visited locations, containing the movement history of the agents, was also used to prevent the agents from repeatedly moving back and forth to the exact location. The experimental results show that this approach performs significantly better but lacks accuracy measures in specific complicated scenarios.

A trusted route planning algorithm for dynamic roads is given in [18]. This work presents an improved trusted model for preventing the wastage of computational resources in path planning. The authors present a Trusted Parallel Optimisation for Planning Dual-Level Grid index methods to attain this objective. This will enable continuous route planning to be done more efficiently. Short-term memory network algorithms are used to predict evolving traffic. The results demonstrate superior performance compared to conventional methods.

A practical implementation of drone technology for conventional paddy fields is given in [19]. In this paper, the authors aim to facilitate the effective implementation of modern technologies in agriculture, thereby enhancing crop yields. The authors address the complexities associated with traditional spraying operations in paddy fields. This is because the sprayer must remain in contact with harmful chemicals for a prolonged period. This work implemented drone technology in spraying pesticides in paddy fields across Asian countries, and the results were observed. The authors utilize Deep Learning (DL) algorithms to enhance the efficiency of the drone system for various agriculture-related activities.

An integrated fuzzy decision-making approach is presented for precision agricultural systems in [20]. This work presents a fuzzy decision-making approach for precision agriculture, based on two significant methods: fuzzy weighted zero-inconsistency and the fuzzy decision by opinion score method. The decision matrix is constructed based on various approaches, including the intersection of payload, dimension criteria, and endurance measures. The results and findings

state that this approach effectively determines the decision criteria with zero inconsistency rate.

Next, the shortest prioritised path-based bidirectional wireless charging method for smart agriculture is evaluated in [21]. This work proposes an innovative solution for charging agricultural sensors. The architecture features a centralised server with energy-constrained devices, enabling reduced energy consumption. Further, it enables efficient data transfer to the centralised server. This approach can be widely used in precision agriculture for optimised resource allocation and management through its enhanced remote monitoring and control features.

IoTaaS: A drone-based IoT-as-a-service framework for smart cities is presented in [22]. The prime focus of this framework is to allow the dynamic provisioning of IoT devices or drones for various services. This process involves a gateway mounted across the drones that provides the required services immediately through distributed cloud services. The proof-of-concept-based model is used to evaluate this approach. From the experiment, it is observed that this approach provides a significant framework for emerging UAV-based applications.

A convolutional neural network (CNN)-based model for texture feature extraction in precision agriculture is given in [23]. The primary objective of this work is to classify leaf diseases. The authors propose a DL-based feature extraction algorithm for identifying and classifying plant diseases. The texture features are extracted using the pre-trained CNN-based classifiers. The publicly available datasets are used for experimentation purposes. From the observation, this approach yields better results than conventional methods, but it requires significant enhancements to deal with emerging, complex situations effectively.

A field evaluation of path planning algorithms in smart farms for autonomous mobile robots is presented in [24]. This work utilises simultaneous localisation and mapping (SLAM) technology for efficient path planning and navigation-related tasks in smart farms. It employs various robots, including spraying, harvesting, and agricultural transport, to perform diverse functions in smart farms. This approach has been experimentally observed to yield better results in identifying static and dynamic obstacles.

An autonomous navigation controller for agricultural vehicles is given in [25]. This approach presents a path-planning method to explore the straight-line reference path using two points and the target path. The navigation controller deals with data processing, information communication, and program execution more efficiently. Various experiments have been conducted to assess the performance of this algorithm. However, this approach lacks robustness and energy-efficient measures due to the inherent design constraints of the underlying architecture.

Kipf and Welling [32] suggested the Semi-Supervised Classification with Graph Convolutional Networks (GCN). An efficient variant of graph-operating CNN is the basis

of our scalable approach to semi-supervised Learning on graph-structured data. To back up our choice of convolutional architecture, the author invokes a localised first-order approximation of spectral graph convolutions. The methodology scales linearly with the number of graph edges to provide hidden layer representations that store both the local network topology and the properties of nodes. Our method surpasses prior methods in several experiments performed on citation networks and a knowledge graph dataset.

Veličković et al. [33] proposed the Graph Attention Networks (GAT). GATs are a new type of neural network that utilises masked self-attentional layers to enhance previous methods, which relied on graph convolutions or their approximations. These GATs process graph-structured input. This enables us to address multiple significant issues simultaneously using a spectral-based GNN, allowing our model to function effectively for inductive and transductive reasoning. Our GAT models have either surpassed or equalled state-of-the-art results on well-known transductive and inductive graph benchmark datasets and one protein-protein interaction dataset (where test graphs are hidden during training).

Liu et al. [34] recommended the Heterogeneous Interaction Graph Transformer (HEIGHT) for Robot Navigation in Constrained and Crowded Environments. Our area of research focuses on robot navigation in highly congested crowds, including obstacles such as hallways and furniture. Earlier approaches yielded inefficient and risky robot routes because they overlooked specific interactions between agents and barriers. To develop robot navigation strategies using deep reinforcement learning (DRL), this paper presents a structured framework with a graph-based description of crowded and limited settings. To begin, the author suggests a heterogeneous spatio-temporal (st) graph to capture the unique interactions between people, robots, and barriers in the environment by splitting the representations of the various components. The author presents HEIGHT, a new navigation policy network architecture based on the heterogeneous st-graph to represent the diverse connections between entities in space and time. The HEIGHT system utilizes attention methods to prioritize critical interactions and employs a recurrent network to monitor the ever-changing landscape, enabling the robot to avoid collisions adaptively. The author demonstrates that HEIGHT outperforms existing baselines in challenging navigational tasks through thorough simulations and real-world testing.

Jin et al. [35] proposed the Inter-Ridge Navigation Path Extraction Method Based on Res2net50 Segmentation Model. Based on the Res2net50 paradigm, this method utilises Squeeze-and-Excitation Networks (SE) to emphasise key visual features. The space pyramid pooling module extracts high-level semantic data and improves fine-grained representation. The retrieved semantic and textural data are combined via a skip connection. Following the ridge prediction, the final picture segmentation is achieved using sampling. The navigation line is fitted after retrieving the

navigation feature points using the ridge segmentation mask. In this paper's inter-ridge navigation route extraction strategy, the Mean Intersection over Union (MIOU) and F-measure values are 0.157 and 0.061 higher than those of the ResNet50 network. Under varying lighting conditions, the average angle error is 1.395° and the pixel error is 8.27 pixels. This method enhances the accuracy of network prediction models for ridge operations.

Fu et al. [36] discussed the Path Planning of Agricultural UAV Based on Improved DRL. Acquiring remote sensing images from Google Earth, segmenting task areas using a DL U-Net model, creating rasterised environmental maps, and calculating coverage paths are the four components of this system. The objective is to make drone pesticide spraying in complicated agricultural settings safer and more efficient. Experiments in simulation demonstrated that, compared to the classic DQN method, the BL-DQN algorithm improved coverage by 41.68%. Compared to the 9.78% reached by the DQN algorithm and the 31.29% achieved by the Depth-First Search (DFS) method, the BL-DQN algorithm had a lower repeat coverage rate of 5.56%. Also, BL-DQN only needed 80.1% as many steps as the DFS algorithm. The BL-DQN algorithm performed even better than DQN and DFS in guiding targets to their exact locations.

Zhang et al. [38] presented the Graph Isomorphism Network with Weighted Multi-Aggregators (GIN-WMA) for building shape classification models. The GIN-WMA utilizes a new aggregator that combines the max and sum aggregators to enhance its identification and discrimination capabilities. Following aggregation using the sum aggregator, this method successfully distinguishes nodes with equal characteristics. The feature extraction process considered local and global form aspects, including influences from Gestalt cognitive psychology and GNN's "node-graph" differentiation method. The impact of different node attributes and combinations on classification accuracy was also investigated, and GIN-WMA's performance was compared to current approaches. The results demonstrated that GIN-WMA is the most effective approach for distinguishing between different building shapes, proving its superiority in shape classification and paving the way for end-to-end building shape extraction and classification.

Moshavash et al. [39] introduced Momentum Contrast (MoCo) Self-Supervised Learning (SSL) for Training Adversarial Robustness. A deep network is pre-trained to perform MoCo as a pre-text job. To make the network more resistant to adversarial assaults, fine-tuning is necessary. The author presents a novel attack approach that utilises MoCo with a non-labelled method, such as Projected Gradient Descent (PGD) or Fast Gradient Sign (FGSM). By pre-training a deep network using corrupted data and an adversarial training approach, we may increase the network's resilience by fine-tuning downstream tasks using the network's representation. In comparison to the Jigsaw, Rotation, and Selfie setups, the one incorporating the ResNet50 structure shows

a 2.79% improvement, the one including the PGD attack shows a 2% improvement, and the one including MoCo-v1 shows a 1.35% improvement. The results section provides further experimental details and MoCo-related improvements, demonstrating that MoCo-based models outperform non-MoCo-based models on CIFAR-10 and CIFAR-10-C datasets.

Grill et al. [40] offered Bootstrap Your Own Latent (BYOL) for SSL. Interacting and learning from one another, BYOL's online and target networks are the neural networks upon which the system is based. The author trains the online network to anticipate the target network's depiction of the same picture under various augmented views by learning from one augmented view to another. Simultaneously, we update the target network using an online network average that moves slowly. Negative pairings are fundamental to current methods, but BYOL reaches a new state of the art without them. Using the conventional linear evaluation technique and a regular ResNet-50 architecture, BYOL achieves 74.3% top-1 classification accuracy on ImageNet. With a larger ResNet, it reaches 79.6%. On both the transfer and semi-supervised benchmarks, we demonstrate that BYOL outperforms the existing.

Liu et al. [41] proposed a traversal path planning and Collaborative Scheduling Approach for Corn Harvesting and Transportation in Hilly Areas, based on the improved Harris Hawk Optimisation (IHHO) and Dijkstra's algorithm. A path-planning and collaborative scheduling technique is presented using Dijkstra's algorithm and IHHO. For full-coverage route planning, a DEM-based field model reduces the length of the traversal path. The IHHO algorithm decreases average scheduling expenses by 4.2% and 14.5%, reduces scheduling time by 4.5% and 8.1%, and decreases fuel usage by 3.5% and 3.2% compared to the classic HHO and HHO algorithms.

Yu et al. [42] proposed the Real-time double-tree rapidly-exploring random tree path planning (RDT-RRT) for autonomous vehicles (AVs). After undergoing B-spline smooth treatment, RRT's collision-free route is used as the reference path to reduce invalid sampling. Integrating G1 and G2 Hermite interpolation decreases sampling dimension and improves efficiency. The ideal distance measure is created for dynamic collision detection and used to evaluate the costs of sample route curvature. A CNN is implemented to improve collision detection and environment awareness. RDT-RRT evaluates smooth, collision-free pathways with modest changes in curvature. Parallel computing is used to test our idea in local and global planning using a real-scale AV. Compared to standard RRT-based algorithms, the solution decreases cumulative curvature change by 92% and 88% in obstacle-free and static obstacle cases, improving route quality.

Zeng et al. [43] suggested the Cost-Efficient GNN Processing Over Distributed Edge Servers. A thorough modelling approach that captures multiple cost considerations is employed to formulate a cost-efficient network

TABLE 1. Summary of the existing methods.

Author(s)	Method/Approach	Key Results	Limitations
Chittoor et al. [16]	Weed detection + Boustrophedon path planning in CowBot	60% path length reduction over traditional methods	Focused on weed mowing; not scalable across different terrain types
Chittoor et al. [17]	Reactive Neural Network for Obstacle Detection	Improved navigation; reduced redundant movements	Accuracy drops in complex obstacle environments
Barburiceanu et al. [18]	Trusted Parallel Optimisation with Grid Indexing	Faster route planning; traffic prediction via STM	Limited scalability in non-road field navigation
Pak et al. [19]	Drone Sprayer with DL Enhancement	Safer pesticide application; improved efficiency in paddy fields	Context-specific (paddy fields); lacks real-time deployment details
Yin et al. [20]	Fuzzy Weighted Zero-Inconsistency Decision-making	Zero inconsistency in decision matrix construction	High dependence on parameter sensitivity
Ab Wahab et al. [21]	Prioritised Bidirectional Wireless Charging	Energy-efficient sensor network; smart agriculture application	Deployment complexity; centralisation limits decentralisation benefits
Xiao et al. [22]	Drone-based IoTaaS Framework	Dynamic provisioning; enhanced service flexibility	Needs robust real-world testing and scaling
Zhao et al. [23]	CNN for Texture Feature Extraction	Effective leaf disease	Struggles with unseen or complex textures

TABLE 1. (Continued.) Summary of the existing methods.

		classification	
Gasparetto et al. [24]	SLAM-based Robot Navigation	Accurate obstacle avoidance; smart farm deployment	Lacks cross-domain testing or generalisation
Yang et al. [25]	Two-Point-Based Navigation Controller	Improved straight-line tracking	Poor energy efficiency; lacks robustness
Kipf & Welling	Graph Convolutional Networks (GCN)	Good semi-supervised learning on graph data	Poor with direction-aware navigation; low spatial dynamics
Veličković et al.	Graph Attention Networks (GAT)	State-of-the-art transductive learning	High computational cost; inefficient on edge devices
Shuijing Liu et al.	HEIGHT (Heterogeneous Interaction Graph Transformer)	Excellent in dense crowds; high safety	High complexity and hardware requirements
Xin Jin et al.	Res2Net50 + SE + SPP for Ridge Navigation	MIOU ↑0.157, F-measure ↑0.061; better under varied lighting	Limited to ridge lines; not generalised
Haitao Fu et al.	BL-DQN + U-Net Segmentation	Coverage ↑41.68%, Repeat ↓5.56%, Steps ↓19.9%	Limited real-world deployment testing
Ya Zhang et al.	GIN-WMA for Shape Classification	Superior classification on building datasets	Focused on shape recognition; indirect to UAV path planning

layout optimisation problem known to be NP-hard. The author theoretically uncovers the structural trait of quadratic

submodularity involved in GNN's distinctive computation pattern, which supports our fast iterative solution via graph

TABLE 1. (Continued.) Summary of the existing methods.

Monireh Moshavash et al.	Momentum Contrast (MoCo) for Robustness	↑2.79% ResNet50 robustness ; better than PGD/FGS M setups	Needs fine-tuning; primarily image classification
Jean-Bastien Grill et al.	BYOL (Bootstrap Your Latent) Own	74.3% Top-1 Accuracy on ImageNet (ResNet-50)	High memory; lacks spatial generalisation
Huanyu Liu et al.	IHHO + Dijkstra for Corn Hilly Harvesting	↓Fuel 3.5%, ↓Scheduling Cost 14.5%, ↑Field Coverage	Region-specific; terrain model required
Jiaxing Yu et al.	RDT-RRT + CNN + B-spline interpolation	↓Curvature Change by 92%; ↑Obstacle avoidance	High computational burden
Liekang Zeng et al.	Cost-Efficient Distributed GNN Processing	95.8% cost reduction, fast convergence	Needs complex edge server architecture
Aye Aye Maw et al.	iADA*-RL (Anytime Dynamic A* + DRL)	High efficiency; tested in AirSim	Simulation-only; lacks real field trials

cuts. It has a parameterized constant approximation ratio, assured convergence, and accurate feasibility, as demonstrated through rigorous analysis. The technique outperforms baselines with more than 95.8% cost reduction and rapid convergence time on real-world database and GNN benchmarks.

Maw et al. [44] proposed the Anytime Graph-Based Path Planning with DRL (iADA*-RL) for an Autonomous UAV. The UAV is deployed in real-life applications after learning about the surroundings in a simulator. Collision avoidance and optimum flight path generation comprise the suggested system. A graph-based route planning algorithm and a learning-based local planning method are combined to enable the UAV to avoid collisions in real-time. A new anytime incremental search method, enhanced Anytime Dynamic A*, solves the global route planning issue in the first stage. To circumvent environmental impediments, local planning between

waypoints uses RL. The hybrid route planning technique was tested in AirSim. AirSim was used for many simulations and tests to test the autonomous UAV system. This paper advances research on UAV route planning algorithms regarding efficiency and safety. Table 1 shows a summary of the existing methods.

A. SUMMARY OF LITERATURE

It is evident from the literature [26], [27], [28], [29], [30] that path planning and navigation in precision farming remain an open issue, and research in this area has been continuously evolving, both historically and currently. Furthermore, a brief review of suitable approaches for path planning has been conducted, and a summary of these approaches is presented in Figure 2. It covers a broader landscape, including path planning, navigation planning, and trajectory control. Finding the most efficient solution remains a complex task.

Furthermore, it imposes various constraints, including optimality, accurate path planning, completeness, cost-efficiency, time efficiency, robustness, energy efficiency, and collision avoidance. Attainment of all these properties requires the most efficient DL algorithm. With this objective, the proposed SGNN presents an enhanced GNN-based prediction algorithm with a self-supervised model for precision agriculture. A detailed view of the proposed architecture is presented in the next section.

Agricultural path-planning challenges often include sparse, unlabeled data organised as irregular topologies. SGNN, which combines the capabilities of graph-based spatial modelling with self-supervised learning, is well-suited to these conditions. SGNN employs learnable propagation functions and dynamic neighbourhood encoding to avoid over-smoothing in deeper layers, unlike GCNs, which use static Laplacian-based aggregation. This allows SGNN to preserve discriminative spatial information over long graph distances. Although GATs provide attention-based adaptive weight assignment, the complexity and instability caused by their softmax-based attention computation become apparent in highly varied agricultural settings, where node degrees vary dramatically. As an alternative, SGNN utilizes localized topological priors to adjust the weight of edges, thereby reducing the computational load without compromising contextual awareness. Graph Transformers and Vision Transformers are examples of transformer-based models. They perform well in dense data regimes but tend to overfit in sparse-field settings, as they require a substantial amount of labelled training data and processing resources. Improving generalizability in the face of real-world limitations, SGNN's self-supervised architecture allows for strong representation learning using unlabeled sensor and trajectory data. In theory, SGNN is a better and more scalable option for innovative agricultural navigation systems due to its topological plasticity, reduced training requirements, and computational efficiency. Figure 3 shows the performance improvement across approaches.

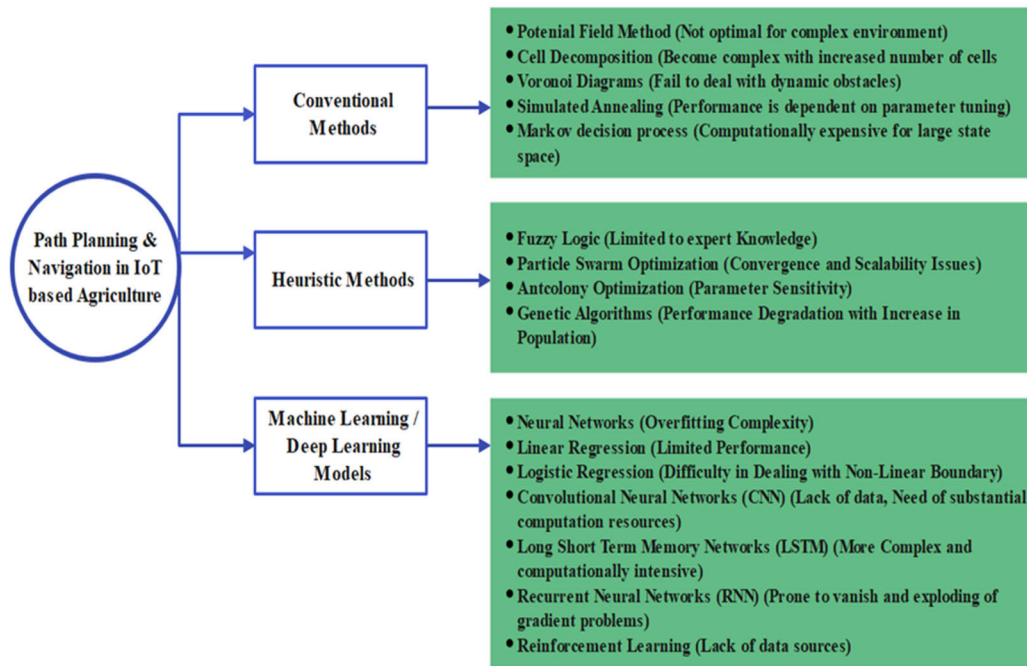


FIGURE 2. Overview of the existing approaches for path planning with their limitations.

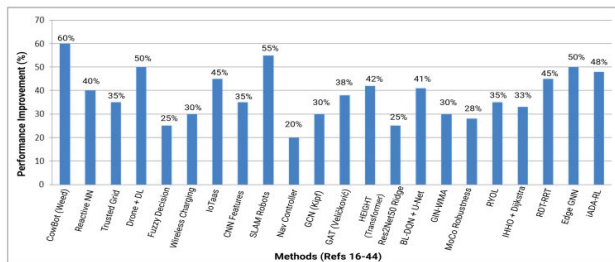


FIGURE 3. Performance improvement across approaches.

Previous research studies are used as benchmarks and comparison points to place the proposed SGNN framework within the current landscape of graph-based and learning-driven path-planning models. Kipf and Welling pioneered the scalable spectral approximation approach for semi-supervised node classification [32] in Graph Convolutional Networks (GCNs), and it was further developed by Veličković et al. [33] in Graph Attention Networks (GATs). GATs use masked self-attention to enhance inductive and transductive reasoning on graph-structured data. Using these as a starting point, Liu et al. [34] demonstrated the advantages of modelling heterogeneous spatio-temporal interactions by proposing the Heterogeneous Interaction Graph Transformer (HEIGHT) to navigate crowded situations. Similarly, Fu et al. [36] achieved considerable improvements in coverage compared to conventional DQN approaches when they used enhanced deep reinforcement learning (DRL) for agricultural route

planning using UAVs. The continuous effort to develop adaptive navigation and job allocation systems of farming robots is shown by other approaches, such as the Res2Net50-based segmentation strategy described by Jin et al. [35] and the IHHO-Dijkstra hybrid scheduling system described by Liu et al. [41]. The self-supervised part of SGNN is based on more generalised findings in self-supervised learning, such as those of MoCo [39] and BYOL [40], demonstrating quantifiable gains in representation learning robustness. It is crucial to optimise computational performance for on-device, real-time applications. Studies such as Zeng et al. [43] on cost-efficient GNN processing at the edge and Maw et al. [44] on DRL-integrated graph-based UAV planning clarify this point. This research provides the overall rationale for the SGNN's hybrid architecture, which combines graph-based spatial reasoning with self-supervised feature learning to provide efficient, generalizable, and resource-aware route planning in diverse agricultural environments.

III. SELF-SUPERVISED LEARNING GRAPHICAL NEURAL NETWORK DRIVEN PREDICTION MODEL(SGNN) FOR SMART SUSTAINABLE AGRICULTURE

The proposed SGNN approach works in two stages. In the first state, arbitrary-shaped regions of interest are modelled using a GAN-based adversarial learning model. This forms the aggregation stage. In the second stage, GNN-based self-supervised learning approaches enhance the quality of the learned representations. Through the careful implementation of GAN algorithms, the initial phase generates a model of small-scale features associated with agricultural field images,

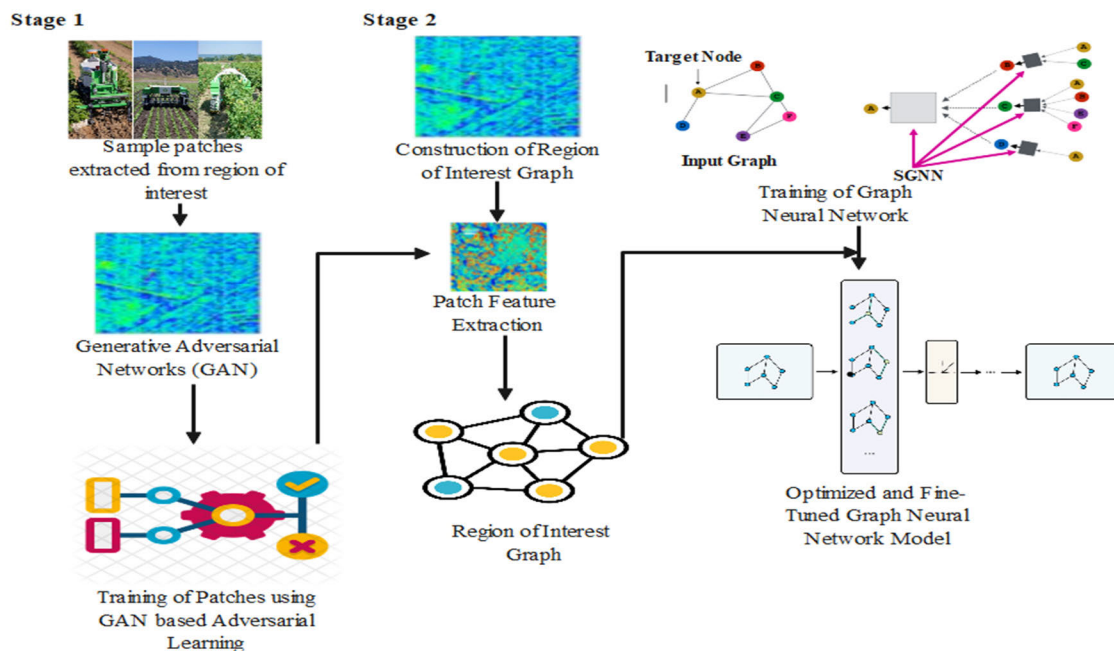


FIGURE 4. Training procedure for the proposed SGNN.

and the second phase models the large-scale structure of these images. The fixed-size patches are extracted from the region of interest during the first stage using the GAN method. During the second stage, the pictures of the area of interest are then constructed into graphs. The trained model obtained from the first stage acts as a patch-level feature extractor. The graph-based neural network model is trained in a self-supervised manner.

This section presents an adversarial training approach for image semantic segmentation. Initially, we construct an adversarial model to verify that the semantic segmentation can be obtained from the ground truth or the segmentation network. The primary objective of the proposed approach is to confirm that it can identify and correct higher-order inconsistencies between the segmentation map produced by the ground truth and the segmentation network. The proposed approach's outline is as follows: Initially, the segmentation network takes an input image and provides per-pixel class predictions. Next, the discriminator takes the maps with labels as input and provides the actual class label (1 for ground truth and 0 for synthetic). The proposed approach trains two model segmentation models and the discriminator model. The ultimate challenge is handling the inherent uncertainty associated with the segmentation task. Here, the variational autoencoders are used for the GAN implementation process. The generative adversarial model is initially constructed using the latent representation L with the simple distribution Q_L . The likelihood $Q_{\Theta}(.|L)$ is constructed using normal distribution with the fixed (learned) parameter σ an.

The neural network outputs the mean value, which is given as input by L . This type of neural network is called a

probabilistic decoder. It maps the latent representation across images, a , and is defined using parameter θ instead.

The Self-Supervised Learning Graphical Neural Network Driven Prediction Model (SGNN) provides an improved method for path planning. This model utilises a graph-based representation of agricultural environments, where elements such as terrains, crop beds, irrigation points, and machinery paths are represented as nodes and edges. Without the need for time-consuming and labour-intensive labelling, SGNN can extract functional characteristics from unlabeled spatial-temporal data, such as images captured by drones, soil sensor readings, and navigation trajectories, all while maintaining the structural context. Accurate prediction of optimum navigation routes even under constantly changing field circumstances is made possible by SGNN's message-passing and neighbourhood aggregation procedures, which capture multi-hop relationships across irregular topologies. In contrast to standard DL models bound by grid-based spatial assumptions, SGNN successfully simulates non-Euclidean geometries and heterogeneous data fusion, supporting real-time adaptation and generalisation across varied farm layouts. SGNN best accomplishes intelligent path planning in smart sustainable agriculture due to its superior route efficiency, reduced computational overhead, and enhanced resilience to field anomalies, such as crop gaps or terrain irregularities. Table 2 presents a comparative justification of SGNN versus other graph-based models for UAV Path Planning.

The objective function is obtained through the variational inference approach, and the maximum likelihood function is estimated using stochastic gradient descent. This, in turn,

TABLE 2. Comparative justification of SGNN vs Other graph-based models for UAV path planning.

Model Type	Spatial Awareness	Edge Feature Utilisation	Handling Irregular Topology	Computational Efficiency	Suitability for UAV Path Planning
Graph Convolutional Network (GCN)	Limited (averages node neighbours)	Fundamental; no dynamic weighting	Moderate	Moderate	Struggles with fine-grained spatial dynamics
Graph Attention Network (GAT)	Moderate (attention on neighbours)	Weights edges via attention	Good	Higher computational overhead	Better than GCN but scales poorly
Graph Transformer	High, global receptive field	Good with learned positional encodings	Excellent	Very high	Accurate but inefficient for real-time UAVs
SGNN (Proposed)	High (uses spatial edge connectivity)	Explicit edge modelling + directional edges	Excellent	Optimised for sparse representations	Balances spatial awareness and efficiency

is used for learning the generative model on larger datasets. Here, q_θ is the incomplete likelihood value for the sample provided $a^{(j)}$. Further, the model is intractable. This is because the computation of this variable needs marginalisation of the complete likelihood across the latent variable.

$$q_\theta(a^{(j)}, x) = \int_x q_\theta(a^{(j)}|x) p_{r_x}(x) dx \quad (1)$$

In addition to the posterior distribution $q_\theta(\cdot|a^{(j)})$, it becomes intractable. Such that, for any distribution P over x , we formulate a tractable lower bound as follows:

$$\begin{aligned} E_{MS}(P(x)||q_\theta(x|a^{(j)})) \\ = \int_x P(x) \log \frac{P(x)}{q_\theta(x|a^{(j)})} dx \\ = \int_x P(x) \log \frac{P(x) q_\theta(a^{(j)})}{q_\theta(a^{(j)}|x) p_x(x)} dx \end{aligned} \quad (2)$$

$$\begin{aligned} = -D_P[\log q_\theta(a^{(j)}|x) + \log q_\theta(a^{(j)})] \\ + E_{MS}(P(L)||q_L(L)) D_P[\log q_\theta(a^{(j)}|x) + \log q_\theta(a^{(j)})] \\ + E_{MS}(P(L)||q_L(L)) \end{aligned} \quad (3)$$

Here, the term E_{MS} represents the value of divergence.

The above equation is further formulated as,

$$\begin{aligned} \log q_\theta(a^{(j)}) - E_{MS}(P(x)||q_\theta(x|a^{(j)})) \\ = E_P[\log q_\theta(a^{(j)}|x)] - E_{MS}(P(x)||q_x(x)) \end{aligned} \quad (4)$$

Here, the divergence value is always non-negative. The lower bound value corresponding to the right-hand side is an incomplete data log likelihood, given as $\log$. This, in turn, is known as the variational lower bound. The variational inference value approximates the distribution P_Φ , and Φ parameterises it. The conventional variational inferences are optimised through the parameters ϕ for the function y . This enhances the efficiency of the inferences. The proposed method selects the approximate posterior based on a Gaussian distribution using diagonal covariance and is represented as $a^{(j)}$. Through this, the network learn for the prediction of mean and covariance using the distribution $P_\phi(\cdot|a^{(j)})$, which further makes an approximation of the posterior.

The objective function is defined as follows:

$$L(\theta, \Phi; a^{(j)}) = E_{P_\Phi}(\log q_\theta(y^{(j)}|x)) - E_{MS}(P_\phi(x|a^{(j)})||q_x(x)). \quad (5)$$

To enable the network to learn about the two modes, the gradient value associated with the minibatch should be estimated using the parameters (θ, ϕ) . Next, the gradient value associated with the second term is computed analytically. But this process is highly complicated. Because the entire distribution is parameterized by ϕ . To overcome this issue, we employ the reparameterization process. Consider a random variable x followed by the normal distribution $N(\mu, \sigma^2 J)$ and given a sample ϵ for a standard normal distribution.

Thus, for any function f we have:

$$E_a[f(a)] = E_\epsilon[f(\mu)\sigma_\epsilon] \quad (6)$$

From the above, we can easily derive the estimator $\frac{1}{S} \sum_{s=1}^S f(\mu)\sigma_\epsilon$ for the first term. Usually, only one sample is used for the estimation. An overview of the training procedure is summarised in Figure 4. Since the model encodes every sample from a minibatch, it provides an approximate posterior with parameters as output, through which the sampling representation is made. Thus, it is referred to as a variational

encoder. It is then later decoded into an appropriate image. Here, the first term denotes the reconstruction loss, which is dependent on the choice of a Gaussian distribution for a particular likelihood. The MS term regularizes the encoder's output.

A. PATCH SAMPLING

Initially, the patches from the region of interest are obtained using two parameters: the patch size and the step size. All patches within a specific area are utilized without requiring additional conditions. At the same time, the other algorithms utilise a fixed algorithm to discard the patches. Further, the region of interest graph defines the representation of the area of interest associated with undirected graphs. Here, the vertices represent the fixed-size patches and the edges are associated with the spatial proximity relation among the patches.

B. SELF-SUPERVISED GNN ALGORITHM

The primary objective of the self-supervised GNN algorithm is to train the GNN encoder using unlabeled data to perform downstream tasks. Based on the pretext task, it is usually categorised as generative, contrastive, or predictive. In the proposed SGNN approach, we utilise the contrastive method.

1) CONTRASTIVE LEARNING

The key idea for a contrastive-based self-supervised learning method is to establish a meaningful concatenation of graphs related to a particular downstream task. The correlated data pairs can discriminate the random pairs by maximising mutual information during augmentation. A brief description of the model is given below:

Implementing a self-supervised GNN model comprises three major components: data augmentation, a contrastive objective, and a contrastive scale. Consider the graph data $G(U, E, P)$, where $\{T_j\}_{j=1}^L$ denotes the data augmentation transformations, and it generates multiple views associated with the graph $\{T_j(H)\}_{j=1}^L = 1$. Then, $\{k_{\theta j}\}_{j=1}^L = 1$ are the pretext encoders that map those views as representations $\{m_1, \dots, m_L\}$.

$$m_j = k_{\theta}(T_j(H)), j = 1, \dots, L \quad (7)$$

The encoders $\{k_{\theta}\}_{j=1}^L = 1$ always share similar parameters, so two augmentations are used here: $L=2$. The task associated with pretext process is formulated as:

$$\theta^*, \Phi^* = \arg \min_{\theta, \Phi} L_{Sesl}(h_{\Phi}(K_{\theta 1}(T_1(H)), \dots, h_{\theta L}(T_L(H)))) \quad (8)$$

Here, the parameter L_{Sesl} represents the estimator that enables the model to maximise the mutual information between positive pairs and they're by minimizing the negative pairs. In this case, the augmentation pairs that belong to the same instances are considered as positive, while the remaining pairs are found to be negative.

During the different stages of data augmentation associated with a model, generating multiple views related to the graph occurs at an initial stage. The augmentation process is closely associated with the graph's topological structure, such that every augmented graph is considered a new graph and shares partial information with other original instances.

$$H_j^{\sim} = (B_j^{\sim}, P_j^{\sim}) = T_j(B, P) = (H)T_j \quad (9)$$

Comparison scale is associated with two or more pretext decoders h_{Φ} . The augmented views across the same scale or various scales are compared using the contrastive methods. The process of comparison between multiple scales enables the model to capture distinct semantic information from the data.

The next module deals with training objectives. The primary intention is to optimise the agreement among various views of the data. Consider a pair of graph representations, m_i, m_j , the agreement that exists among the representations using random variables is computed using the mutual information.

$$U_J(g_i, g_j) = E_x(g_i)_x(g_j) \quad (10)$$

2) GENERATIVE MODELS

The key focus of the generative model is to strengthen the connection between the hidden space mapped by the encoder and the space associated with the target component. The working of the proposed self-supervised algorithm is illustrated in Figure 5.

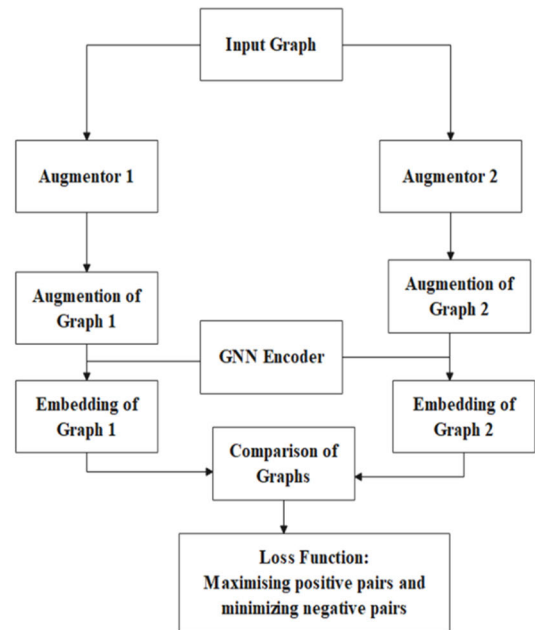


FIGURE 5. Visualisation of Self-Supervised GNN.

3) DESCRIPTION OF THE MODEL

The major modules associated with this model include a model extractor, an objective loss function and a decoder

extractor h_ϕ . Let us consider the input data $h(v, E, P)$; particular components associated with H are extracted from the target component t_H . Then the component is reconstructed from the graph embedding made by GNN-encoder K_θ using the decoder h_ϕ .

$$S_H^{\sim} = h_\phi(k_\theta(H)) \quad (11)$$

The distance between the target component and the concerned reconstructed version is minimised using the objective loss function L_{sesl} .

$$\theta^*, \phi^* = \arg_{\theta, \phi} \min_{\phi} L_{sesl}(h_\phi(k_\theta(H)), S_H) \quad (12)$$

Here, the significant difference between self-supervised graph learning and other approaches is considered the target component in the proposed approach. The smallest subset of graph edges/node attributes, or the adjacency matrix, is regarded as the generic components extracted for reconstruction. The encoder facilitates this process learning the global structure and local connections. Once the reconstructed target is specified for meaningful learning, the encoder k_θ must accept the input graph without any target information. In the case of attribute reconstruction, masking or noising is used to modify the features partially. During the rebuilding of the adjacency matrix, particular ratios of the edges are randomly perturbed because of the noisy topology.

Thus, a particular part S_H associated with G_H is selected for the reconstruction process using the component extractor for a given input data G . In this case, the GNN encoder algorithm is used to map the slightly modified version of the S_G to the embeddings of g_H . After this process, the decoder h_ϕ is reconstructed from G_H to rebuild the target component S_H . Here, L_{sesl} reduces the distance between $S_G^{\sim}$ and S_G .

In general, the decoder h_ϕ is usually designed to prevent the acquisition of larger amounts of information from the data. This property can diminish the learning feature associated with the encoder k_θ . Here, the loss function L_{sesl} is described using the Euclidean distance between the original and reconstructed ones.

Next, the attribute masking algorithm is used to train the GNN encoder, which is achieved by reconstructing node and edge attributes. The encoder K_θ helps mask the attributes mapped to the vectors. Then the linear classifier h_ϕ is used to predict target attributes using corresponding embeddings.

a: EDGE PREDICTION

The objective of the edge prediction algorithm is to reconstruct the adjacency matrix from the node embeddings. The proposed approach utilizes the graph embedding approach, an extension of POI recommendation. The primary objective is to minimise the log-loss value by predicting the context using instance embedding as an input feature. The proposed algorithm uses a similar approach, but here we adopt a negative sampling approach to adapt the characteristics of the heterogeneous graph well.

In the proposed SGNN approach, the heterogeneous graph is divided into three bipartite graphs (user graph, activity

graph, and time graph). Initially, we define the graph embedding method for negative sampling.

b: EMBEDDING OF BIPARTITE GRAPH

Let us consider a bipartite graph $H_{PQ}(U_P, V_Q, E)$. Here, the two disjoint sets are represented using U_P and U_Q , and the edges between them are defined. The objective is to find the parameter θ concerning the model $A_\theta(U_i|U_j)(U_i \in U_P)$ context vertex. $U_j \in U_Q$ is the target vertex. This enables the approximation of empirical distribution value $A^{\sim}(U_i|U_j)$ through cross-entropy (CE) minimisation. Thus, the empirical distribution value associated with the graph is given as:

$$A^{\sim}(U_i|U_j) = \frac{F_{ij}}{\deg(j)} \quad (13)$$

Here, F_{ij} denotes the edge weight between the vertices U_i and U_j or zero, if the two vertices are not connected.

U_i gives the conditional probability measure associated with the vertex, which is generated as an output of softmax by U_j .

$$A_\theta(u_i|u_j) = \frac{e^{w_i^{-s} z_j^{-s}}}{\sum_{i' \in U_P} e^{z_{i'}^{-s} z_j^{-s}}} \quad (14)$$

The embedding associated with the vertex is U_u , represented as Wu^{-s} . Thus, for every vertex U_j in U_Q , a condition distribution $q(\cdot|U_j)$ exists across all the set's vertices. Further, for every pair of vertices U_i, U_j , the computation of the second-order proximity is determined using the conditional distributions $A_\theta(\cdot|U_j), A_\theta(\cdot|U_i)$.

To find the embedding that can assure the conditional distribution $A_\theta(\cdot|U_j)$ can be closely approximated through empirical distribution $A^{\sim}(\cdot|U_j)$. This in turn is minimized using the objective function across the graph G_{PQ} :

$$M_{PQ} = \sum_{j \in U_Q} \lambda_j l(A^{\sim}(\cdot|U_j), a_\theta(\cdot|U_j)) \quad (15)$$

The KL-divergence between two distributions are given as $l(\cdot, \cdot)$. Here, λ_j denotes the significance of the vertex U_j in the graph. Finally, the objective function is represented by replacing $l(\cdot, \cdot)$. Here λ_j denotes the importance of the vertex U_j in the graph. Finally, the objective function is represented by replacing $l(\cdot, \cdot)$ with the KL-divergence by fixing $\lambda_j = \deg(j) = \sum_{j \in U_P} Z_{ij}$ and deleting some constant values.

$$M_{PQ} = -\sum_{(i,j) \in E} Z_{ij} \log a_\theta(U_i|U_j) \quad (16)$$

c: NEGATIVE SAMPLING

The objective's optimisation process is computationally expensive, and it also needs summation operation over the entire set of vertices during the computation of conditional probability $a_\theta(\cdot|U_j)$. To resolve this issue, the proposed method employs the negative sampling technique, rather than considering every pair of nodes; a smaller set of observed edges is sampled. Then, the various negative edges for each observed edge are sampled negatively. To be more specific, the binary edge $e(i, j)$ is sampled based on the likelihood measure, which is proportional to their weight Z_{ij} , and then,

Algorithm 1 Training algorithm for Embedding Bipartite Graph

Input: bipartite graph (user graph, time graph, activity graph); vector dimension v_d , negative sample v and number of iterations n .

Output: Latent node embedding for users (robots) ($w_v \in C^{|v| \times v_d}$, activities: $w_{p \in c}$ and $w_{t \in c}$).

Course of Action:

TRAIN ($n, v, V_d, H_{qu}, H_{qt}, H_{p,q}$).

Initialise W_v, W_q, W_t and W_p

While iteration $\leq N$ do

BPG $E(H_{qv}, u)^k$ (Update W_p, W_v)

BPG $E(H_{qu}, u)^k$ (Update W_q, W_t)

BPG $E(H_{pq}, u)^k$ (Update W_p, W_t)

Return W_v, W_q, W_t and W_p

Set of Instructions: BPGE(H_{PQ}, V)

Edge sampling $e_{ij}(u_i \in u_p, u_j \in u_Q)$

Sample K negative nodes from $B(\cdot | j)$ (represented as u'_j)

Update W_p, W_q , and W'_p

using a particular noise distribution $B(i')$, multiple negative edges (i, j) are sampled. Thus, $B(i)^\alpha \deg(i)^{3/4}$. Meanwhile, $\deg(i)$ represents the degree of vertex U_i . This indicates the most popular vertices are more likely to be chosen from the chosen negative samples. This makes sense when the adjacency matrix associated with the graph is highly sparse. The primary objective of negative sampling is to distinguish between the context path/vertex and another path/vertex that is more likely to be a context. The proposed negative sampling method is more efficient as it replaces the noise distribution $b(i)$ over the alternative $b(i/j)$. The target vertex is represented as U_j and U_i

Produces the negative sample vertex.

$$b(i/j) \propto 1 - \frac{z_{ij}}{d(j)} \times \text{Acf}(i) \quad (17)$$

Through the use of the edge sample and negative sample discussed above, the last objective function associated with the bipartite graph G_{pq} is given as:

$$M_{PQ} = -\sum_{(i,j) \in E} [\log \sigma(w_i^{->T} w_j^{->}) + \sum_{d=1}^v \Xi_{u_i'}(b(\cdot | j)) \cdot (\log \sigma(-w_i^{->T} w_j^{->})] \quad (18)$$

In the above equation, σ denotes the sigmoid function, and then V negative instances are sampled for every positive example. The edge e_{ij} is sampled during every iteration based on the gradient value. So that the embedding vector $w_i^{->}$ that corresponds to the vertex, it is computed as $\frac{\delta M_{PQ}}{\delta w_i}$.

d: TRAINING ALGORITHM ASSOCIATED WITH EMBEDDINGS OF BIPARTITE GRAPH

Once the model learns the embeddings using the user graph, the time graph and the activity graph. Then these embeddings

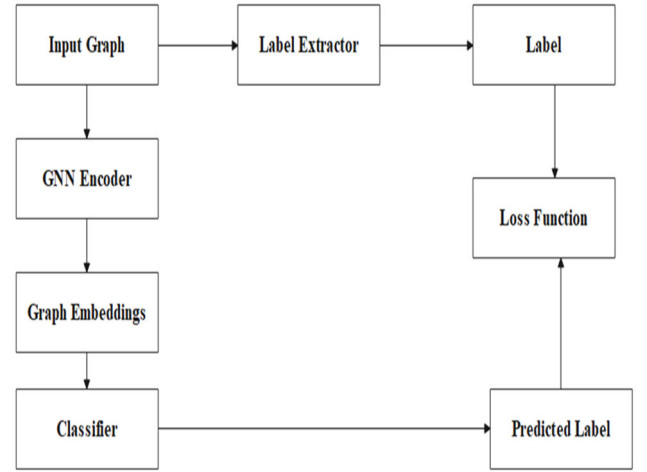


FIGURE 6. Visualisation of Predictive GNN model.

are used to predict and suggest a potential path for a robot through pairwise similarity. Here, the user graph represents the path associated with the robots, activity denotes the activity performed by the robot, and the time graph indicates the estimated time for the robots to move from one place to another. When two or more swarm robots wish to move towards a particular location, then the robot $v, \forall u \in V \setminus v$, we calculate $w_v^T W_v$ and the results are ranked among robots V , the set of robots in navigation. Through this process, the top-ranked robot will be more likely to move towards a particular path.

4) PREDICTIVE MODELS

Using a predictive model facilitates the efficient generation of informative labels by leveraging the specific properties of unlabeled data. Then, through the use of pseudo labels, supervised tasks are performed. In the case of self-supervised GNN, there exist three central modules: label extractor le , label classifier h_ϕ and L_{sesl} , the objective function G . For a given input graph $G(U, E, P)$, the pseudo-label q_G is generated using the label extractor le for the given graph G . Meanwhile, the embeddings $h_G = K_\theta(G)$ are mapped using the GNN encoder K_θ , Such that the label q_G is classified as:

$$q_G = le(g) \quad (19)$$

$$q_G = h_\phi(K_\theta(G)) \quad (20)$$

The use of L_{sesl} maximized the prediction accuracy:

$$\theta^*, \phi^* = \arg_{\theta, \phi} \min L_{sesl}(h_\phi(K_\theta(G)), le(G)) \quad (21)$$

Here, the label extractor l is generated based on graph information in the proposed approach. The label extractor is designed reliably to enhance performance using self-supervised learning with lower collection costs: Classifier h_ϕ and objective L_{sesl} . Here, h_ϕ denotes the simple neural network, which adjusts the dimension across data embedding to label size.

The predictive model of SSL-GNN is given in Figure 6. It states that for a particular input graph G , the label extractor compiles the label q_G associated with it. The GNN encoder g_θ connects and maps G with embedding h_G . The label q_G is predicted from h_G using the classifier h_ϕ . And finally, the function L_{ssl} is used to enhance the prediction accuracy. There is a label or category assigned to every pixel in the picture. Through pixel-level annotation, semantic picture segmentation enables ML models to make accurate predictions and gain a deeper understanding of the visual input, thereby facilitating precise object localisations and boundary delineation. The forecast results guide the route planning process to enhance the algorithm's performance.

In this manner, the proposed algorithm assists in more efficient and robust path planning activities in precision agriculture.

IV. EXPERIMENTAL OBSERVATIONS AND DISCUSSIONS

A. MATERIALS AND METHODS

The aerial semantic segmentation dataset [31] is taken for experimental purposes and loaded as a CSV file. The input file contains various classes of images; every class has its associated objects. Every object is given a distinct colour to differentiate between different kinds of objects. The first and foremost step in training our semantic segmentation model is to prepare the dataset. This is because a high understanding of the images is required to perform the path planning process. The proposed algorithm should effectively identify various objects in a real-time environment and their corresponding pixels. The RGB values corresponding to each label are illustrated in Table 4. The names displayed in the table are inserted across the list datatype to count the number of classes available in the dataset. This information will later be used to train the ML model.

This research utilized the Hyperspectral Library of Agricultural Crops (USGS) [37], accessible through Kaggle, to further validate the SGNN model. The USGS contains 6,988 high-dimensional pixel spectra of five main U.S. crops: winter wheat, rice, corn, soybeans, and cotton. It also includes crop growth stage labels for six phenological phases: emergence, vegetative, critical, maturation, senescence, and harvest. This dataset, collected by NASA's Hyperion sensor throughout the United States from 2008 to 2015, contains spectra, geographic coordinates, and agroecological zone information. It is possible to test SGNN's adaptability to variations in spectral signatures based on crop type, growth stage, and regional differences. Table 3 shows the dataset comparison. Figure 7 shows the normalised comparison of dataset attributes.

Once the dataset is loaded, each class is assigned a unique identity, representing the corresponding pixel's class label. The segmentation images are generated and placed across the training and testing dataset folders in the next step. The arbitrarily shaped region of interest is initially modelled using the GAN algorithm. Since the proposed approach works in two

TABLE 3. Dataset comparison.

Data set Name	Data Type	Crop Types	Growth Stage Info	Spatial Coverage	Temporal Coverage	Spectral Bands	Geo-referenced
Semantic Drone Dataset	RGB Image + Semantic Labels	Urban vegetation + ground features	Not Available	Singl e urban site	Not specified	3 (RGB)	No
USGS Hyperspectral Dataset	Hyperspectral Signatures	Winter wheat, rice, corn, soy beans, cotton	6 stages (e.g., vegetative to harvest)	Multiple agroecological zones in the U.S.	2008–2015	200+ (Hyperion sensor)	Yes

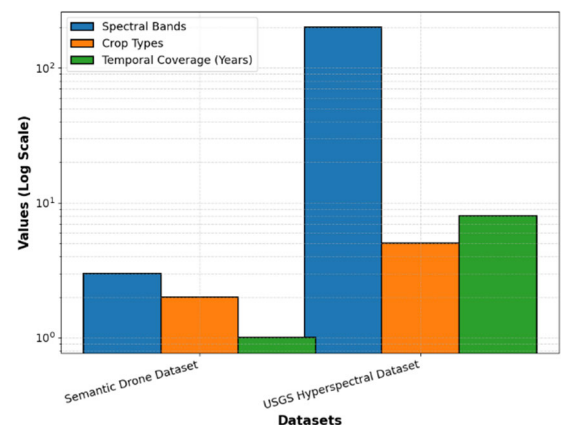


FIGURE 7. Normalised comparison of dataset attributes.

stages. In the first stage, small-scale features associated with the field are extracted. Then these features are utilized to build an effective path prediction model using a self-supervised graph neural network. The set of masked images generated during the first stage is represented in Figure 8.

TABLE 4. RGB values corresponding to each class label [31].

S. No	Name	R	G	B
0	unlabeled	0	0	0
1	paved-area	128	64	128
2	Dirt	130	76	0
3	Grass	0	102	0
4	Gravel	112	103	87
5	Water	28	42	168
6	Rocks	48	41	30
7	Pool	0	50	89
8	vegetation	107	142	35
9	Roof	70	70	70
10	Wall	102	102	156
11	Window	254	228	12
12	Door	254	148	12
13	Fence	190	153	153
14	fence-pole	153	153	153
15	Person	255	22	96
16	Dog	102	51	0
17	Car	9	143	150
18	Bicycle	119	11	32
19	Tree	51	51	0
20	bald-tree	190	250	190
21	ar-marker	112	150	146
22	obstacle	2	135	115
23	conflicting	255	0	0

Next, the model is trained to make segmentations. Here, we compare the original image, the predicted image, and the annotated image to assess the performance of the proposed model during its training progression. The proposed method provides good validation accuracy at the end of the training phase. The overall accuracy measures associated with the system will be discussed in detail later in this section. The overview of the predicted images at the end.

The training phase is illustrated in Figure 9.

TABLE 5. Dataset 1 (Loss per epoch – Semantic Drone Dataset).

Epoch	Training Loss	Validation Loss
0	2.7	1.7
1	1.6	1.2
2	1.2	1.2
3	1.2	1.3
4	1.4	1.6
5	1	0.95
6	0.9	0.85
7	0.9	0.82
8	0.85	0.8
9	0.8	0.78
10	0.75	0.74
11	0.7	0.7
12	0.68	0.68
13	0.66	0.66
14	0.65	0.64

Next, the trained model is validated using a new set of images. The sample images from the validation phase are shown in Figure 10. Finally, the suggested SGNN method is compared with the proper and forecast masks. The resultant images are briefly illustrated in Figure 11.

B. DISCUSSIONS

Some of the experimental observations that are interpreted during the experiment are discussed in detail throughout this section. After the successful implementation of the proposed model, the loss function and accuracy measures are evaluated to assess the performance. Initially, the loss function associated with the model per epoch is analyzed. The results show that the proposed algorithm has a comparatively reduced loss function during the validation phase compared to the training phase. The loss function gradually decreases as the number of epochs increases. Finally, during the last epoch, the loss function associated with the training and validation epochs gradually reaches a value of less than 1%. On average, the loss function during the training phase is around 1.4%, and during the testing phase, it is 0.4%. An overview of the loss function measures associated with each epoch is clearly illustrated in Figure 11. The primary objective of assessing the loss function parameter is to determine how effectively the model performs with various parameter values.

Figure 13 shows the loss per epoch measure analysis on dataset 2. Tables 5 and 6 show the loss per epoch for datasets 1 and 2.

Next, the mean intersection over union (mIoU) measures are analysed, and the results are illustrated in Figure 14. The mIoU score provides the mean value associated with each class across the dataset. This metric is used to predict the accuracy of the segmentation measures. The proposed SGNN approach provides better mIoU measures in the training and testing phases. On average, the proposed model achieves a score of 0.20, which is a comparatively good result.

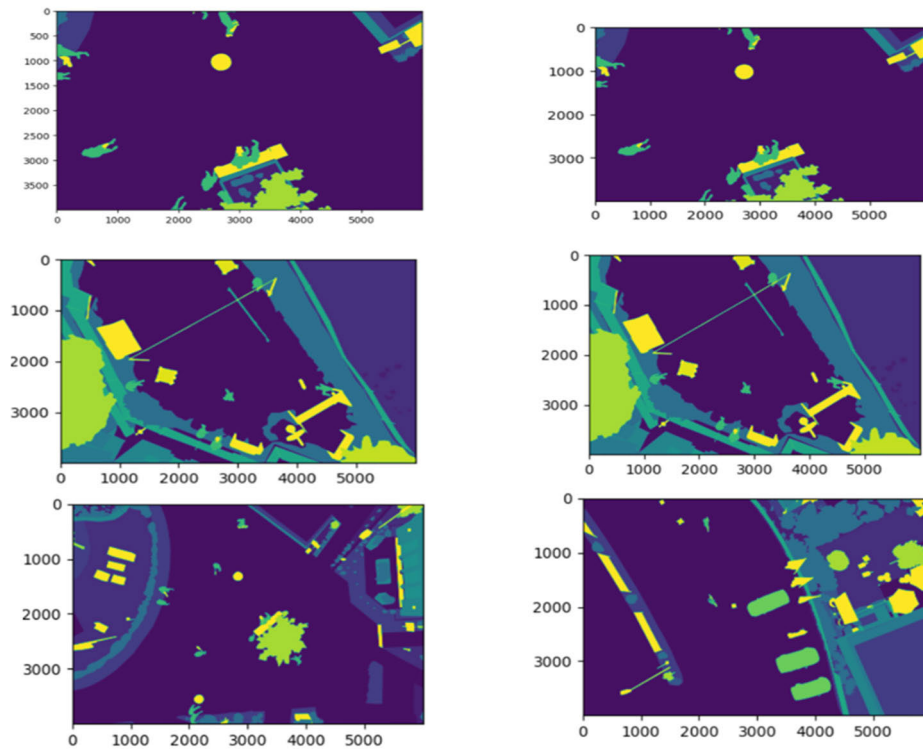


FIGURE 8. Sample pairs of labelled images that are extracted from original images.

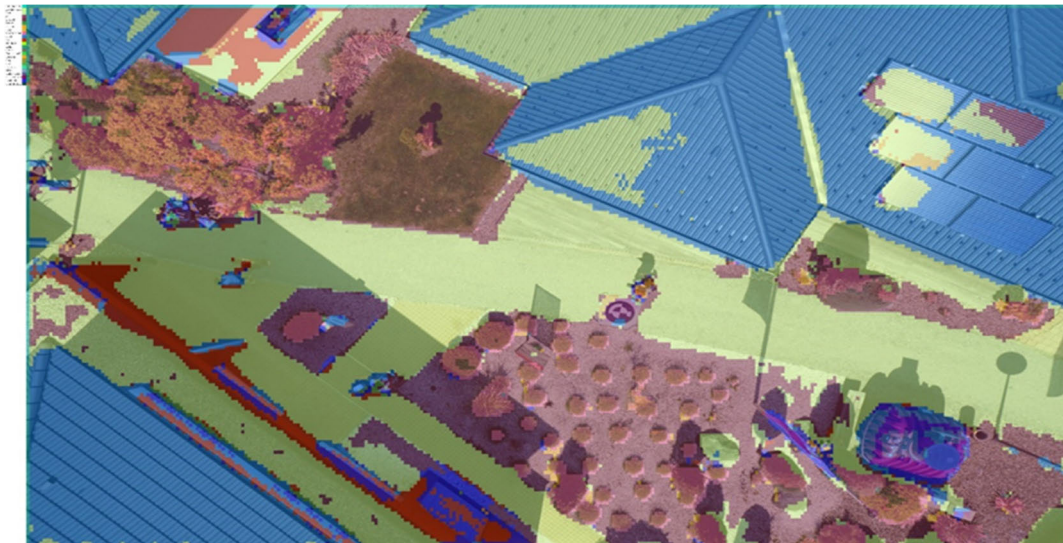


FIGURE 9. Overview of the predicted images at the end of the training phase.

Figure 15 shows the analysis of the MIOU measure on dataset 2. Tables 7 and 8 show the MIOU measure on datasets 1 and 2.

Next, the accuracy measure related to the proposed algorithm is analyzed in detail. The accuracy measures are compared during both the training and testing phases. The experiment reveals that the proposed model achieves an accuracy score of 0.85, equivalent to approximately 85%,

which is a commendable result. The use of a GNN algorithm enables the accurate prediction of measures. Additionally, with the increase in epochs, the accuracy measures significantly improve, yielding better precision. The results are depicted in Figure 16.

Figure 17 shows the comparison of classification accuracy measures on dataset 2. Tables 9 and 10 show the accuracy per epoch for datasets 1 and 2.

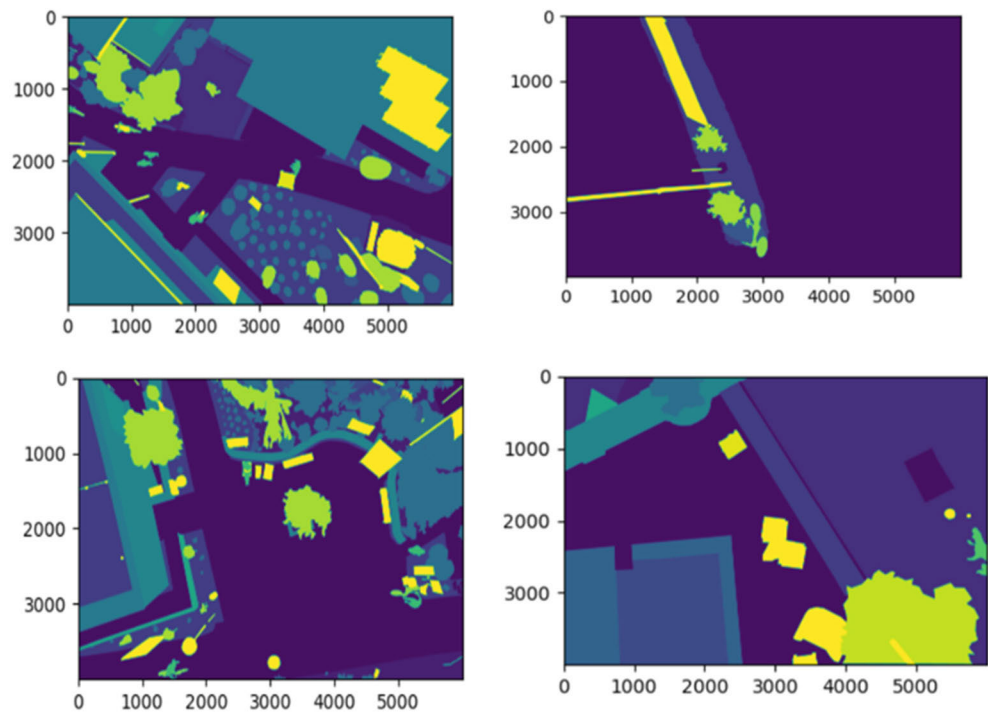


FIGURE 10. Predicted mask images of the validation data.

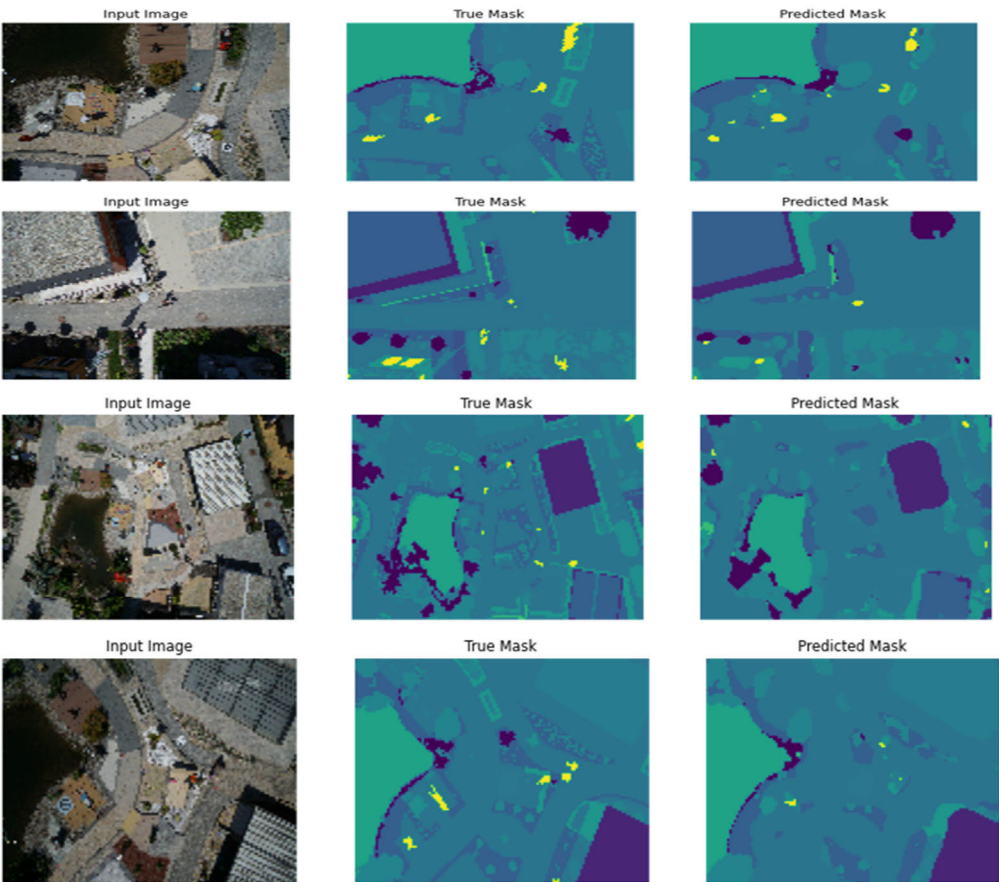


FIGURE 11. Illustration of the true mask and predicted mask for the given input image.

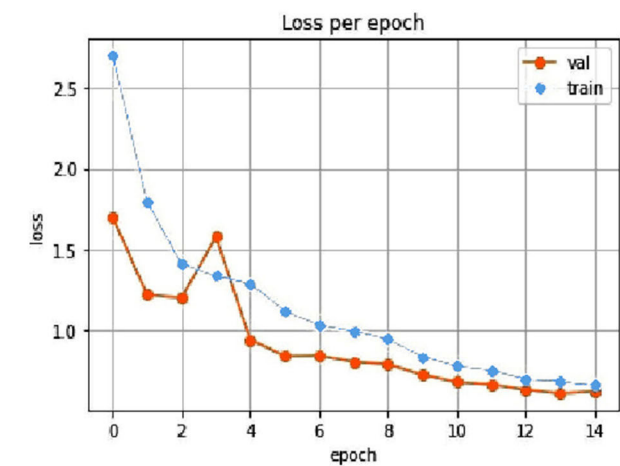


FIGURE 12. Analysis of loss per epoch measure on Dataset 1.

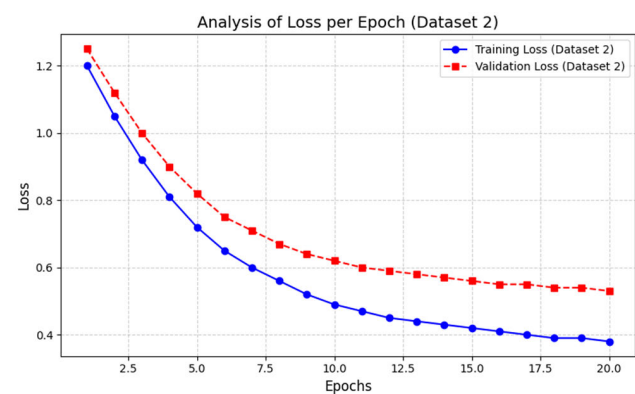


FIGURE 13. Analysis of loss per epoch measure on Dataset 2.

TABLE 6. Dataset 2 (Loss per epoch – USGS Hyperspectral Dataset).

Epoch	Training Loss	Validation Loss
1	1.18	1.24
2	1.12	1.2
3	1	1.1
4	0.88	0.95
5	0.78	0.87
6	0.7	0.8
7	0.64	0.73
8	0.58	0.68
9	0.54	0.64
10	0.5	0.61
11	0.48	0.59
12	0.46	0.58
13	0.45	0.57
14	0.44	0.57

Next, the confusion matrix associated with the proposed method is computed. The confusion matrix defines the segmentation accuracy between the ground truth and the predicted labels. The confusion matrix indicates that out of 939 pixels, 150 are correctly classified as obstacles, such as grass or a person, and 45 samples are misclassified as

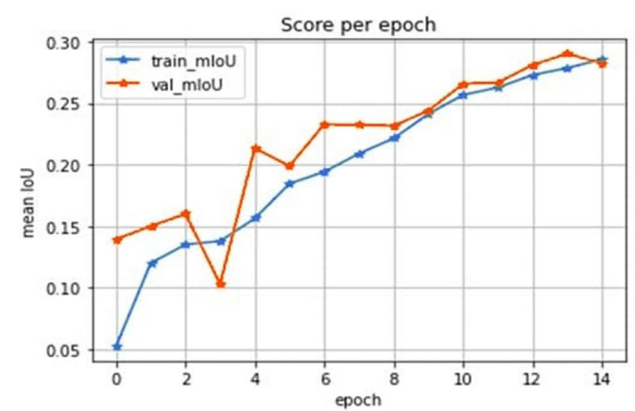


FIGURE 14. Analysis of MIOU measure on Dataset 1.

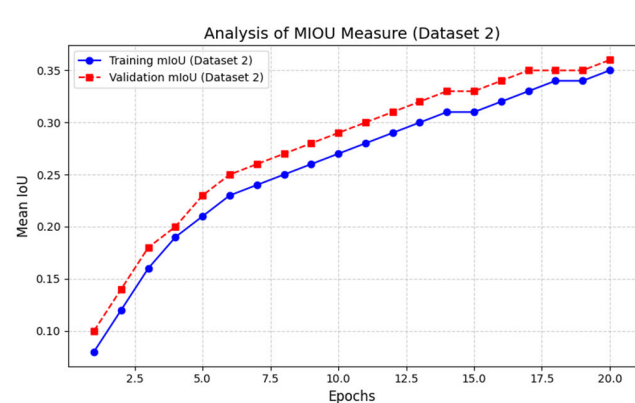


FIGURE 15. Analysis of MIOU measure on Dataset 2.

TABLE 7. Dataset 1 (MIOU measure – Semantic Drone Dataset).

Epoch	Train mIoU	Validation mIoU
0	0.05	0.14
1	0.12	0.16
2	0.13	0.17
3	0.14	0.1
4	0.16	0.22
5	0.19	0.2
6	0.2	0.23
7	0.21	0.23
8	0.23	0.24
9	0.24	0.25
10	0.26	0.26
11	0.27	0.26
12	0.28	0.28
13	0.29	0.29
14	0.28	0.29

obstacles. Based on the confusion matrix, various other performance-related parameters are also analyzed, and the results are illustrated in Table 5. The experiment reveals that the proposed model achieves better classification accuracy

TABLE 8. Dataset 2 (mIoU measure – USGS Hyperspectral Dataset).

Epoch	Train mIoU	Validation mIoU
0	0.05	0.14
1	0.12	0.16
2	0.13	0.17
3	0.14	0.1
4	0.16	0.22
5	0.19	0.2
6	0.2	0.23
7	0.21	0.23
8	0.23	0.24
9	0.24	0.25
10	0.26	0.26
11	0.27	0.26
12	0.28	0.28
13	0.29	0.29
14	0.28	0.29

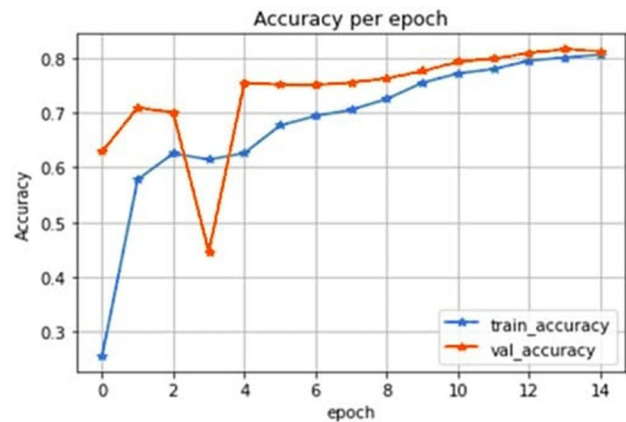


FIGURE 16. Comparison of classification accuracy measures on Dataset 1.

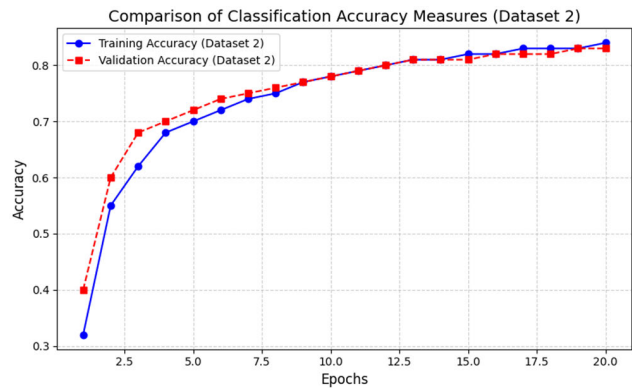


FIGURE 17. Comparison of classification accuracy measures on Dataset 2.

for the segmented images, making it the most suitable model for real-time implementation.

The following experiment assesses the robustness of the proposed algorithm compared to standard conventional

TABLE 9. Dataset 1 (Accuracy per epoch – Semantic Drone Dataset).

Epoch	Training Accuracy	Validation Accuracy
0	0.25	0.63
1	0.59	0.71
2	0.62	0.7
3	0.63	0.45
4	0.67	0.75
5	0.69	0.75
6	0.7	0.76
7	0.72	0.77
8	0.74	0.78
9	0.76	0.79
10	0.78	0.8
11	0.79	0.81
12	0.8	0.81
13	0.81	0.81
14	0.81	0.81

TABLE 10. Dataset 2 (Accuracy per epoch – USGS Hyperspectral Dataset).

Epoch	Training Accuracy	Validation Accuracy
0	0.32	0.54
1	0.55	0.68
2	0.62	0.7
3	0.69	0.73
4	0.72	0.75
5	0.74	0.77
6	0.76	0.78
7	0.78	0.79
8	0.79	0.8
9	0.8	0.81
10	0.81	0.81
11	0.82	0.82
12	0.83	0.82
13	0.83	0.83
14	0.84	0.83

architectures available for computer vision and object detection. From the observation, in a few instances, the proposed algorithm fails to maintain the robustness factors due to various reasons, such as noise, blur, and image compression. This algorithm could be further developed to provide even better results, even in noise and image compression. A comparative analysis of the proposed method and standard architectures is presented in Tables 11 and 12. The CE measures associated with the standard models are reimplemented and assessed in Table 4. The table presents a detailed assessment of the CE measure for the proposed self-neural network model based on mean-corruption and relative mean-corruption errors. The results are then compared with those of standard models, as described in Table 13. According to the observation, the proposed algorithm yields a better relative

TABLE 11. Performance analysis of proposed SGNN method.

Measure	Value
Sensitivity	0.8895
Specificity	0.8384
Precision	0.8907
Negative Predictive Value	0.8609
False Positive Rate	0.8616
False Discovery Rate	0.7093
False Negative Rate	0.2105
Accuracy	0.8586
F1 Score	0.7893
Matthews Correlation Coefficient	0.6858

mean-corruption error compared to other approaches. The proposed algorithm exhibits a lower error corruption rate across various corruptions.

The SGNN model can adapt to different crop varieties, soil structures, and weather-induced fluctuations. It improves generalisation across different agricultural contexts by combining topological reasoning with self-supervised learning. The model can distinguish between variations in spatial row alignment, canopy shape, and inter-crop spacing by analysing data from several crop datasets, including wheat, soybeans, maize, and cotton. Graph node and edge attributes record soil heterogeneity, including changes in compaction, texture, and drainage patterns. This allows the planner to adjust route selection according to topographical restrictions. By utilizing temporal augmentation and node-level feature propagation, the model can simulate weather conditions such as cloudy illumination, wet field conditions, and seasonal variations. This enables the model to maintain its navigation accuracy under varying visibility and terrain conditions. Cross-dataset validation using 5-fold splits further demonstrated low variance in performance metrics (standard deviation $\leq 3.2\%$), confirming the robustness of SGNN's generalization to real-world variability in smart sustainable agriculture contexts.

Using a combination of weighted loss functions and data-level augmentation methodologies, we addressed the problem of class imbalance in the training dataset, specifically the imbalance between obstacle and non-obstacle classes. To make the model more sensitive to sparse and irregular obstacle patterns, a class-weighted cross-entropy loss was used to punish misclassification of minority obstacle locations more harshly. The training set's obstacle representation was further enhanced using data augmentation methods, including synthetic obstacle injection, spatial flipping, and obstacle-centric cropping.

C. SCALABILITY AND REAL-WORLD FEASIBILITY ANALYSIS

Figure 18 compares the SGNN model's performance on the NVIDIA Jetson Nano and Raspberry Pi 4B platforms.

TABLE 12. Top-1/Top-5 accuracy result comparison.

Loss	Architecture	Augmentation Method	Top-1	Top-5
CE (Cross-Entropy) (standard)	Res-Net 50	Mix-Up	75.4	93
CE (standard)	Res-Net 50	Cut-Mix	78.5	93.8
CE (standard)	Res-Net 50	Auto-Augment	78.1	92.1
CE (Proposed)	Graph-Net	Auto-Augment	79.1	94.8
CE (Cross-Entropy) (standard)	Res-Net 200	Auto-Augment	80.4	94
CE (standard)	Res-Net 200	Stacked-Random Augment	80.3	94.8
CE (standard)	Res-Net 200	Stacked-Random Augment	80.1	95.1
CE (Proposed)	Graph-Net	Stacked-Random Augment	83.1	95.9

TABLE 13. Comparative analysis of mean-corruption and relative mean-corruption error rates.

Loss Value	Architecture	Rel. MCE	MCE
CE (standard)	Alex Net	100	99
CE (standard)	Res-Net 18	132	89.2
CE (standard)	Res-Net 50	104.7	85.2
CE (Proposed)	Res-Net 50	95.4	67.2
CE (Proposed)	Res-Net 200	70.5	51
Self-Supervised Graph Network	Graph-Net	95.6	68.2

It displays the average inference time (milliseconds) and memory usage (gigabytes). The results show that SGNN is effective on all devices, indicating its suitability for real-time UAV-based agricultural applications.

Its computational efficiency and scalability were evaluated to assess the practical deployment potential of the proposed Self-Supervised Learning Graphical Neural Network Driven Prediction Model (SGNN) in smart sustainable agriculture. We utilised an NVIDIA Jetson Nano edge computing device

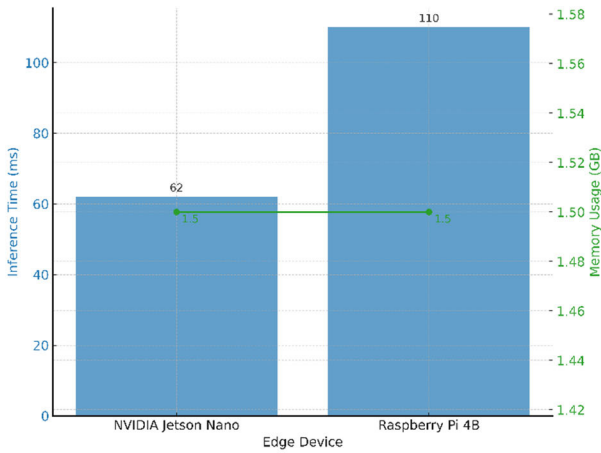


FIGURE 18. Runtime performance comparison of SGNN on edge devices.

TABLE 14. Scalability comparison of edge devices based on inference time and memory usage.

Edge Device	Inference Time (ms)	Memory Usage (GB)
NVIDIA Jetson Nano	62	1.5
Raspberry Pi 4B	110	1.5

with 4GB of RAM, a standard specification in UAV and robot platforms, to execute and evaluate the SGNN model. The evaluations indicated that the model requires approximately 1.5 GB of RAM for operation and takes an average of 62 ms to process each input sample, demonstrating that it can work almost in real-time for activities such as path planning and navigation. Additionally, the model's graph-based design exhibits linear scalability, as the computational complexity increases linearly with the number of agents' neighbors, rather than in a diagonal line. This means that it can deploy multiple agents in a swarm without significantly dropping performance. Additional experiments on a Raspberry Pi 4B with 8GB of RAM confirmed that it was possible, with an inference time of approximately 110 ms, which remains suitable for low-frequency navigation updates in agricultural applications. These results demonstrate that the SGNN model does not require significant processing power or memory. It is ideal for real-time deployment on UAVs and edge devices in smart agriculture. The model can be combined with hardware-in-the-loop field testing to examine energy use, battery life, and its performance in various circumstances. This will help demonstrate that it can be effectively applied in real-world agricultural environments on a large scale. Table 14 presents a scalability comparison of Edge Devices based on Inference Time and Memory Usage.

D. COMPARATIVE EVALUATION OF SGNN AGAINST GRAPH-BASED AND TRANSFORMER-BASED MODELS

According to the comparison results, SGNN is the most effective and accurate model currently available, surpassing

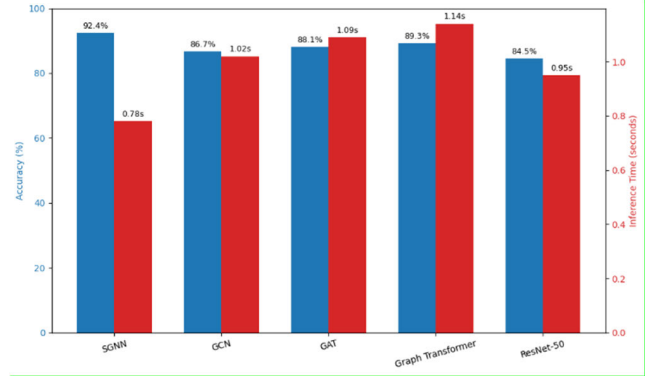


FIGURE 19. Comparison of SGNN with baseline models (GCN, GAT, Graph Transformer, and ResNet-50) in terms of accuracy (%) and inference time (seconds).

ResNet-50, Graph Transformer, GCN, and GAT. With an accuracy of 92.4%, SGNN outperforms all other networks in capturing the contextual characteristics and spatial relationships crucial for route planning in agricultural settings. While both GCN and GAT are graph-based, they exhibit lower accuracy (86.7% vs. 88.1%, respectively) due to issues such as fixed neighbourhood aggregation and increased computational cost resulting from attention methods. Graph Transformers outperform GCN and GAT with an accuracy of 89.3%; however, their increased inference time due to dense self-attention calculations makes them unsuitable for real-time applications. The failure of ResNet-50, a convolutional baseline, to simulate the non-Euclidean spatial connections present in field topologies results in its lowest accuracy of 84.5%. SGNN's computational efficiency and real-time application in innovative sustainable agricultural systems are validated by its shortest inference time of 0.78 seconds, significantly lower than the inference times of other graph-based models, which exceed 1 second. Figure 19 compares SGNN with baseline models, including GCT, GAT, graph transformer, and ResNet-50, regarding accuracy and inference time.

To further validate the robustness of the proposed SGNN architecture, we performed a 5-fold cross-validation analysis on both the Semantic Drone Dataset and the USGS Hyperspectral Dataset. The performance measures showed consistently low variance, with standard deviations rarely exceeding 3.2% among folds. The 95% confidence intervals (CIs) were calculated as follows: accuracy = $85.8\% \pm 2.1\%$, F1-score = 0.789 ± 0.018 , and Matthews correlation coefficient (MCC) = 0.686 ± 0.021 . These narrow CIs show that the results are trustworthy. Also, a paired t-test showed that SGNN's benefits over baseline topologies like ResNet-50, GAT, and Graph Transformer were statistically significant at $p < 0.05$. These results additionally prove that SGNN can make accurate predictions, however it additionally demonstrates that it can be used multiple times and is statistically valid. This makes it even better for use in smart, sustainable agriculture in the real world.

V. CONCLUSION

The application of drone technology in precision agriculture is expected to transform its potential in several ways. Specifically, it can even lead to the attainment of the ultimate goal related to precision agriculture, which includes fulfilling society's current needs with a focus on food without causing any environmental destruction. The proposed SGNN approach presents a promising solution for sustainable farming. Using a GNN algorithm with self-supervised learning enables the significant movement of swarm robots with a reduced likelihood of collision and enhanced path planning capabilities. We perform the semantic segmentation of the agricultural field images for efficient path planning and navigation. The experiments are conducted using the Semantic Drone Dataset, and the results are analysed in detail from various perspectives. Initially, the performance of the proposed method is validated individually. Then, a comparative analysis is made using standard approaches. The experiment reveals that the proposed model achieves more than 85% accuracy across various obstacle detection scenarios, a significantly better score than standard approaches. Furthermore, the proposed model is more robust and effective than traditional approaches.

REFERENCES

- [1] D. Li, B. Li, H. Feng, S. Kang, J. Wang, and Z. Wei, "Low-altitude remote sensing-based global 3D path planning for precision navigation of agriculture vehicles—beyond crop row detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 210, pp. 25–38, Apr. 2024.
- [2] R. F. Carpio, C. Potena, J. Maiolini, G. Ulivi, N. B. Rosselló, E. Garone, and A. Gasparri, "A navigation architecture for Ackermann vehicles in precision farming," *IEEE Robot. Autom. Lett.*, vol. 5, no. 2, pp. 1103–1110, Apr. 2020.
- [3] S. Chakraborty, D. Elangovan, P. L. Govindarajan, M. F. ELnaggar, M. M. Alrashed, and S. Kamel, "A comprehensive review of path planning for agricultural ground robots," *Sustainability*, vol. 14, no. 15, p. 9156, Jul. 2022.
- [4] V. Moysiadis, N. Tsolakis, D. Katikaridis, C. G. Sørensen, S. Pearson, and D. Bochtis, "Mobile robotics in agricultural operations: A narrative review on planning aspects," *Appl. Sci.*, vol. 10, no. 10, p. 3453, May 2020.
- [5] A. T. Meshram, A. V. Vanalkar, K. B. Kalambe, and A. M. Badar, "Pesticide spraying robot for precision agriculture: A categorical literature review and future trends," *J. Field Robot.*, vol. 39, no. 2, pp. 153–171, Mar. 2022.
- [6] J. A. Oroko and G. N. Nyakoe, "Obstacle avoidance and path planning schemes for autonomous navigation of a mobile robot: A review," in *Proc. Sustain. Res. Innov. Conf.*, 2022, pp. 1–16.
- [7] G. Nalinipriya, B. Baluswarny, R. Patan, S. Kallam, T. Gs, and M. R. Babu, "To detect and recognize object from videos for computer vision by parallel approach using deep learning," in *Proc. Int. Conf. Adv. Comput. Commun. Eng. (ICACCE)*, Paris, France, Jun. 2018, pp. 336–341, doi: 10.1109/ICACCE.2018.8441718.
- [8] S. Campbell, N. O'Mahony, A. Carvalho, L. Krpalkova, D. Riordan, and J. Walsh, "Path planning techniques for mobile robots a review," in *Proc. 6th Int. Conf. Mechatronics Robot. Eng. (ICMRE)*, Feb. 2020, pp. 12–16.
- [9] L.-S. Liu, J.-F. Lin, J.-X. Yao, D.-W. He, J.-S. Zheng, J. Huang, and P. Shi, "Path planning for smart car based on Dijkstra algorithm and dynamic window approach," *Wireless Commun. Mobile Comput.*, vol. 2021, no. 1, pp. 1–12, Jan. 2021.
- [10] J. Liu, J. Yang, H. Liu, X. Tian, and M. Gao, "An improved ant colony algorithm for robot path planning," *Soft Comput.*, vol. 21, no. 2017, pp. 5829–5839.
- [11] P. Maini, B. M. Gonultas, and V. Isler, "Online coverage planning for an autonomous weed mowing robot with curvature constraints," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 5445–5452, Apr. 2022, doi: 10.1109/LRA.2022.3154006.
- [12] A. Kanezaki, J. Nitta, and Y. Sasaki, "GOSELO: Goal-directed obstacle and self-location map for robot navigation using reactive neural networks," *IEEE Robot. Autom. Lett.*, vol. 3, no. 2, pp. 696–703, Apr. 2018, doi: 10.1109/LRA.2017.2783400.
- [13] B. Li, T. Dai, W. Chen, X. Song, Y. Zang, Z. Huang, Q. Lin, and K. Cai, "T-PORP: A trusted parallel route planning model on dynamic road networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 1238–1250, Jan. 2023, doi: 10.1109/TITS.2022.3216310.
- [14] S. D. Panjaitan, Y. S. K. Dewi, M. I. Hendri, R. A. Wicaksono, and H. Priyatman, "A drone technology implementation approach to conventional paddy fields application," *IEEE Access*, vol. 10, pp. 120650–120658, 2022, doi: 10.1109/ACCESS.2022.3221188.
- [15] A. N. Jasim, L. C. Fourati, and O. S. Albahri, "Evaluation of unmanned aerial vehicles for precision agriculture based on integrated fuzzy decision-making approach," *IEEE Access*, vol. 11, pp. 75037–75062, 2023, doi: 10.1109/ACCESS.2023.3294094.
- [16] P. K. Chittoor, B. Chokkalingam, R. Verma, and L. Mihet-Popa, "An assessment of shortest prioritized path-based bidirectional wire-less charging approach toward smart agriculture," *IEEE Access*, vol. 11, pp. 123742–123755, 2023, doi: 10.1109/ACCESS.2023.3329976.
- [17] M. A. Hoque, M. Hossain, S. Noor, S. M. R. Islam, and R. Hasan, "IoTaaS: Drone-based Internet of Things as a service framework for smart cities," *IEEE Internet Things J.*, vol. 9, no. 14, pp. 12425–12439, Jul. 2022, doi: 10.1109/JIOT.2021.3137362.
- [18] S. Barburiceanu, S. Meza, B. Orza, R. Malutan, and R. Terebes, "Convolutional neural networks for texture feature extraction. Applications to leaf disease classification in precision agriculture," *IEEE Access*, vol. 9, pp. 160085–160103, 2021, doi: 10.1109/ACCESS.2021.3131002.
- [19] J. Pak, J. Kim, Y. Park, and H. I. Son, "Field evaluation of path-planning algorithms for autonomous mobile robot in smart farms," *IEEE Access*, vol. 10, pp. 60253–60266, 2022, doi: 10.1109/ACCESS.2022.3181131.
- [20] X. Yin, Y. Wang, Y. Chen, C. Jin, and J. Du, "Development of autonomous navigation controller for agricultural vehicles," *Int. J. Agricult. Biol. Eng.*, vol. 13, no. 4, pp. 70–76, 2020.
- [21] M. N. A. Wahab, S. Nefti-Meziani, and A. Atyabi, "A comparative review on mobile robot path planning: Classical or meta-heuristic methods?" *Annu. Rev. Control*, vol. 50, pp. 233–252, Jan. 2020.
- [22] X. Xiao, B. Liu, G. Warnell, and P. Stone, "Motion planning and control for mobile robot navigation using machine learning: A survey," *Auto. Robots*, vol. 46, no. 5, pp. 569–597, Jun. 2022.
- [23] S. Zhao and S.-H. Hwang, "Complete coverage path planning scheme for autonomous navigation ROS-based robots," *ICT Exp.*, vol. 10, no. 1, pp. 83–89, Feb. 2024.
- [24] A. Gasparetto, P. Boscariol, A. Lanzutti, and R. Vidoni, "Path planning and trajectory planning algorithms: A general overview," in *Motion and Operation Planning of Robotic Systems: Background and Practical Approaches*. Cham, Switzerland: Springer, 2015, pp. 3–27.
- [25] L. Yang, J. Qi, D. Song, J. Xiao, J. Han, and Y. Xia, "Survey of robot 3D path planning algorithms," *J. Control Sci. Eng.*, vol. 2016, pp. 1–22, Apr. 2016.
- [26] M. M. Costa and M. F. Silva, "A survey on path planning algorithms for mobile robots," in *Proc. IEEE Int. Conf. Auto. Robot Syst. Competitions (ICARSC)*, Apr. 2019, pp. 1–7.
- [27] H. Noborio, T. Naniwa, and S. Arimoto, "A quadtree-based path-planning algorithm for a mobile robot," *J. Robotic Syst.*, vol. 7, no. 4, pp. 555–574, Aug. 1990.
- [28] M. N. Zafar and J. C. Mohanta, "Methodology for path planning and optimization of mobile robots: A review," *Proc. Comput. Sci.*, vol. 133, pp. 141–152, Jan. 2018.
- [29] L. Liu, X. Wang, X. Yang, H. Liu, J. Li, and P. Wang, "Path planning techniques for mobile robots: Review and prospect," *Expert Syst. Appl.*, vol. 227, Oct. 2023, Art. no. 120254.
- [30] T. F. Abaas and A. H. Shabeeb, "Path planning optimization of a mobile robot based on intelligence algorithm," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 881, no. 1, Jul. 2020, Art. no. 012047.
- [31] *Aerial Semantic Segmentation Drone Dataset Downloaded From Kaggle*. Accessed: Feb. 20, 2025. [Online]. Available: <https://www.tugraz.at/index.php?id=22387>
- [32] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.
- [33] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, and Y. Bengio, "Graph attention networks," 2017, *arXiv:1710.10903*.

- [34] S. Liu, H. Xia, F. C. Pouria, K. Hong, N. Chakraborty, Z. Hu, J. Biswas, and K. Driggs-Campbell, "HEIGHT: Heterogeneous interaction graph transformer for robot navigation in crowded and constrained environments," 2024, *arXiv:2411.12150*.
- [35] X. Jin, C. Lin, J. Ji, W. Li, B. Zhang, and H. Suo, "An inter-ridge navigation path extraction method based on Res2net50 segmentation model," *Agriculture*, vol. 13, no. 4, p. 881, Apr. 2023.
- [36] H. Fu, Z. Li, W. Zhang, Y. Feng, L. Zhu, X. Fang, and J. Li, "Research on path planning of agricultural UAV based on improved deep reinforcement learning," *Agronomy*, vol. 14, no. 11, p. 2669, Nov. 2024.
- [37] Accessed: Feb. 20, 2025. [Online]. Available: https://www.kaggle.com/datasets/billbasener/hyperspectral-library-of-agricultural-crops-usgs?utm_source=chatgpt.com
- [38] Y. Zhang, J. Liu, Y. Wang, Y. Cao, S. Xu, and A. Luo, "Graph isomorphism network with weighted multi-aggregators for building shape classification," *Trans. GIS*, vol. 28, no. 6, pp. 1883–1904, Sep. 2024.
- [39] M. Moshavash, M. Eftekhari, and K. Bahraman, "Momentum contrast self-supervised based training for adversarial robustness," *J. Comput. Secur.*, vol. 8, no. 1, pp. 33–43.
- [40] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. A. Pires, Z. Guo, M. G. Azar, B. Piot, k. Kavukcuoglu, R. Munos, and M. Valko, "Bootstrap your own latent—A new approach to self-supervised learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21271–21284.
- [41] H. Liu, J. Luo, L. Zhang, H. Yu, X. Liu, and S. Wang, "Research on traversal path planning and collaborative scheduling for corn harvesting and transportation in hilly areas based on Dijkstra's algorithm and improved Harris hawk optimization," *Agriculture*, vol. 15, no. 3, p. 233, Jan. 2025.
- [42] J. Yu, C. Chen, A. Arab, J. Yi, X. Pei, and X. Guo, "RDT-RRT: Real-time double-tree rapidly-exploring random tree path planning for autonomous vehicles," *Expert Syst. Appl.*, vol. 240, Apr. 2024, Art. no. 122510.
- [43] L. Zeng, C. Yang, P. Huang, Z. Zhou, S. Yu, and X. Chen, "GNN at the edge: Cost-efficient graph neural network processing over distributed edge servers," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 3, pp. 720–739, Mar. 2023.
- [44] A. A. Maw, M. Tyan, T. A. Nguyen, and J.-W. Lee, "iADA-RL: Any-time graph-based path planning with deep reinforcement learning for an autonomous UAV," *Appl. Sci.*, vol. 11, no. 9, p. 3948, Apr. 2021.



R. NIVETHA received the Bachelor of Engineering degree in electronics and communication engineering from VIIT, in 2015, and the Master of Engineering degree in applied electronics from Kingston Engineering College, Tamil Nadu, India, in 2017. She is currently pursuing the Ph.D. degree with the School of Electronics Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu. Her research interests include path planning and localization of robot, automations, edge computing, and the Internet of Things.



K. C. SRIHARIPRIYA is currently pursuing the Doctor of Philosophy degree with Anna University. She is a Postdoctoral Fellow with IIT Madras, in indigenous 5G testbed project. She has a strong belief in the principle of experiential learning and versatile approach with sensitivity to individual learning needs. She has 18 years of teaching and research experience. She has co-authored the book "Handbook of Cognitive Radio" in 2019. She has been a 3GPP RAN 1 delegate representing IIT Madras, working toward the standards and specifications for 5G in 2019, supported by TSDSI, DoT, Government of India. She has filed patents on Intelligent Communication for Smart Grid (Application No.: 202041032691 A) and Portable Real-time Cardiac Anomalies Classification and Analysis System Using Tiny ML (Application No.: 202441012592). She has delivered multiple Guest lectures and hands on Training programs on cognitive radio, 5G and beyond, the Internet of Things (IoT), machine learning, multi-disciplinary IoT applications, Python, MATLAB, simulation tools for IoT, and data analytics, Workshops using Arduino and Raspberry Pi. Her research interests include cognitive radio, communication and signal processing, 5G and beyond, artificial intelligence and machine learning for medical applications, deep learning, the IoT, and data analytics.



BALAMURUGAN BALUSAMY received the bachelor's, master's, and Ph.D. degrees in India. He is currently the Associate Dean of Academics with the Shiv Nadar Institution of Eminence Deemed to be University, New Delhi. Prior to this assignment, he was a Professor with the School of Computing Sciences and Engineering and the Director of International Relations of Galgotias University, Greater Noida, India. His contributions focus on engineering education, block chain, and data sciences. His academic degrees and 12 years of experience working as a Faculty Member in global universities have made him more receptive and prominent in his domain. He does have 200+ high impact factor papers in Springer, Elsevier, and IEEE. He has done more than 200 edited and authored books and collaborated with eminent professors across the world from top QS ranked universities. His passion is teaching and adapting different design thinking principles while delivering his lectures. He has visited 15+ countries for his technical course. He serves in the advisory committee for several startups and forums, and does consultancy work for the industry on Industrial IoT. He has given over 195 talks in various events and symposium. He has spearheaded and conducted more than ten IEEE and ACM International conferences up to date.

...